

BFS Traversal

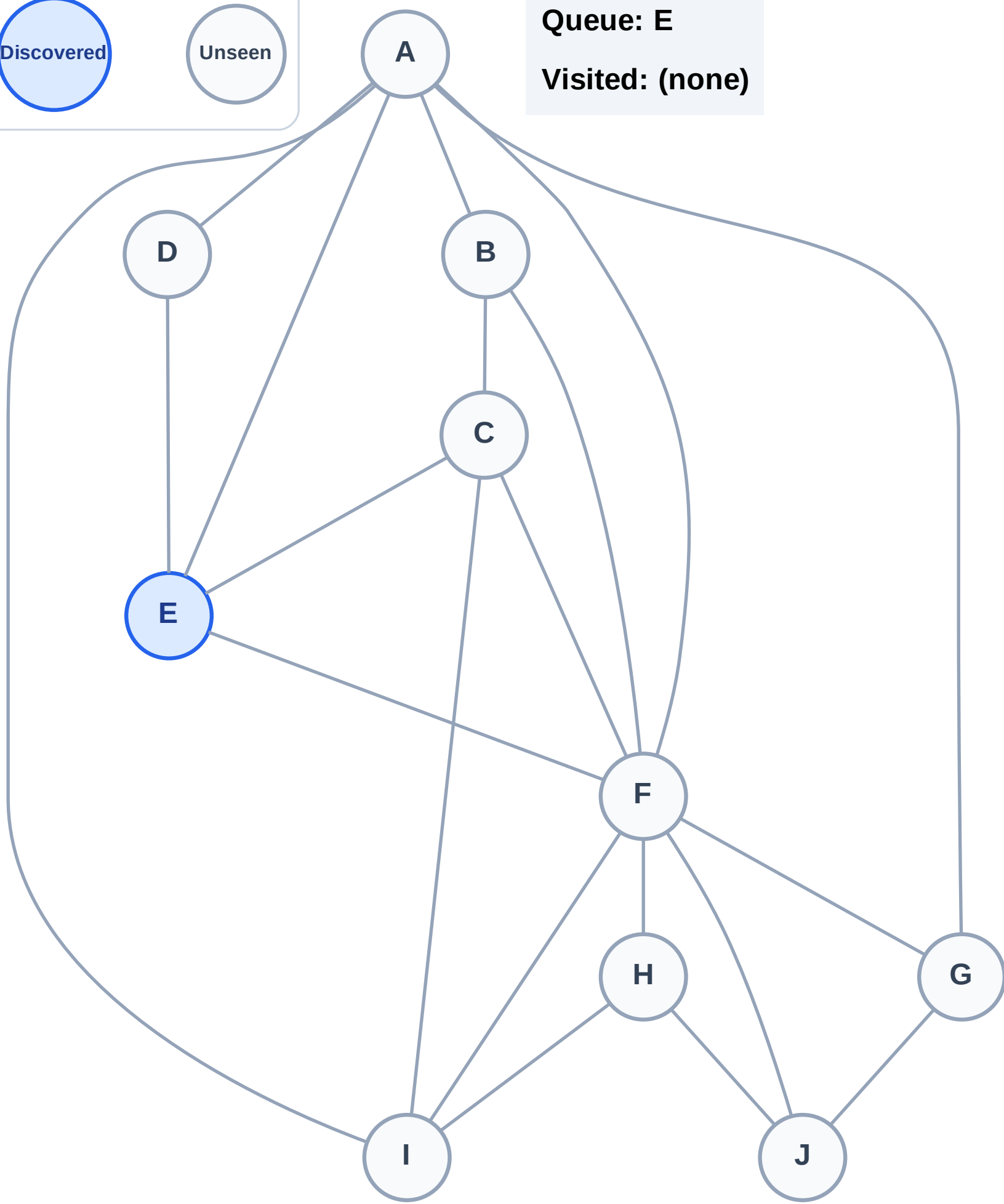
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

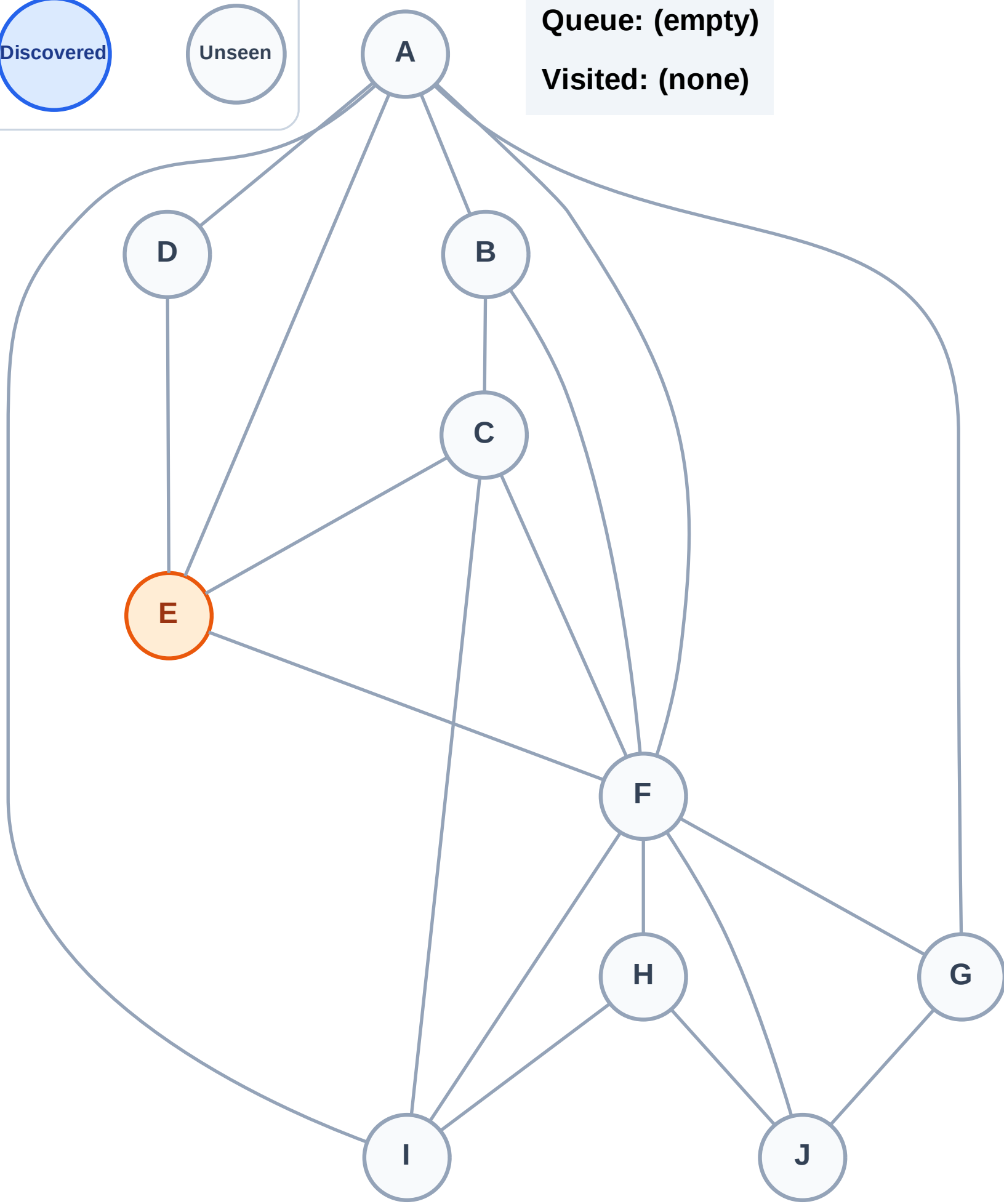
Step 2: Process node E

Legend



Queue: (empty)

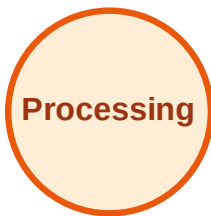
Visited: (none)



BFS Traversal

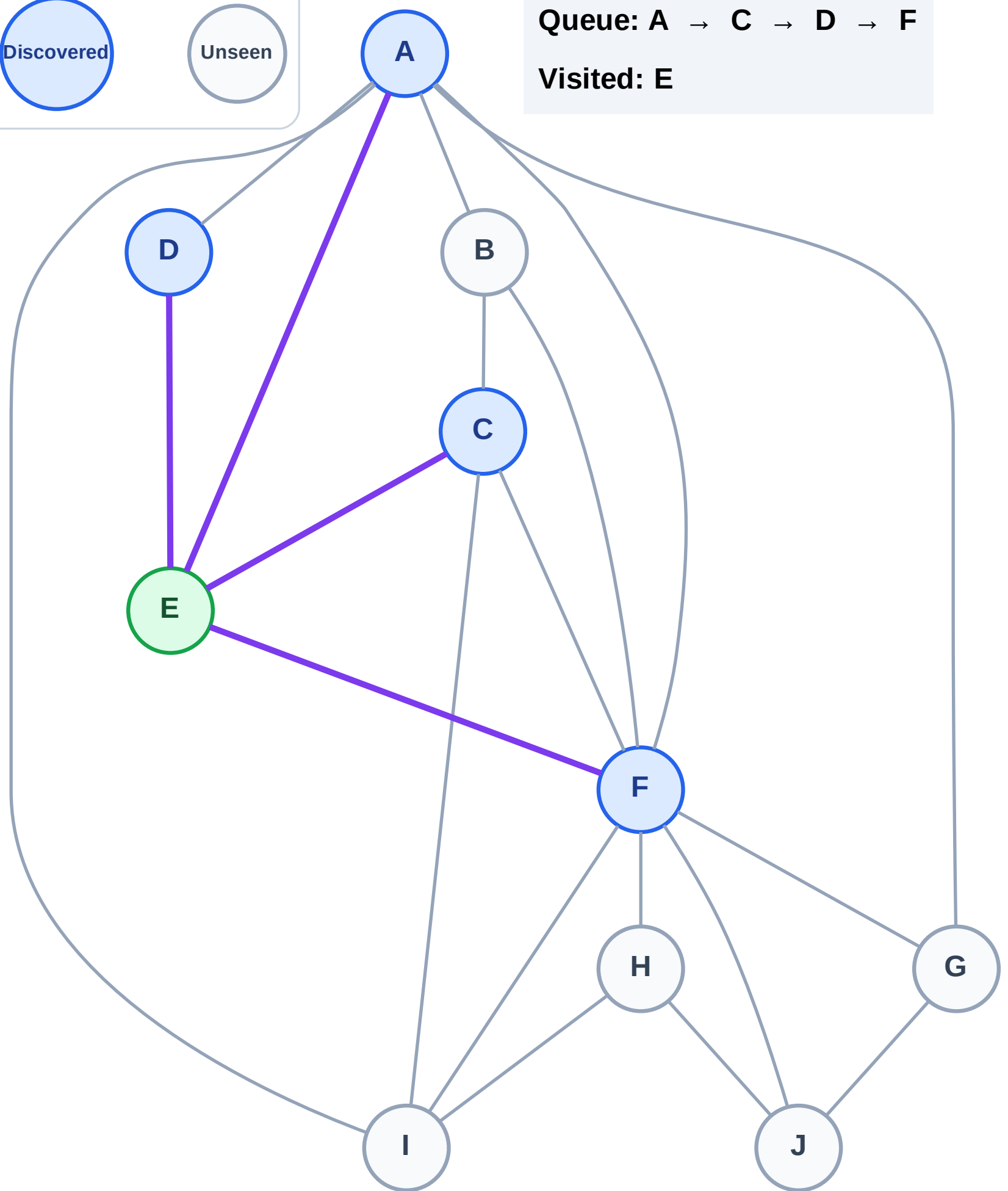
Step 3: Discover A, C, D, F from E

Legend



Queue: A → C → D → F

Visited: E



BFS Traversal

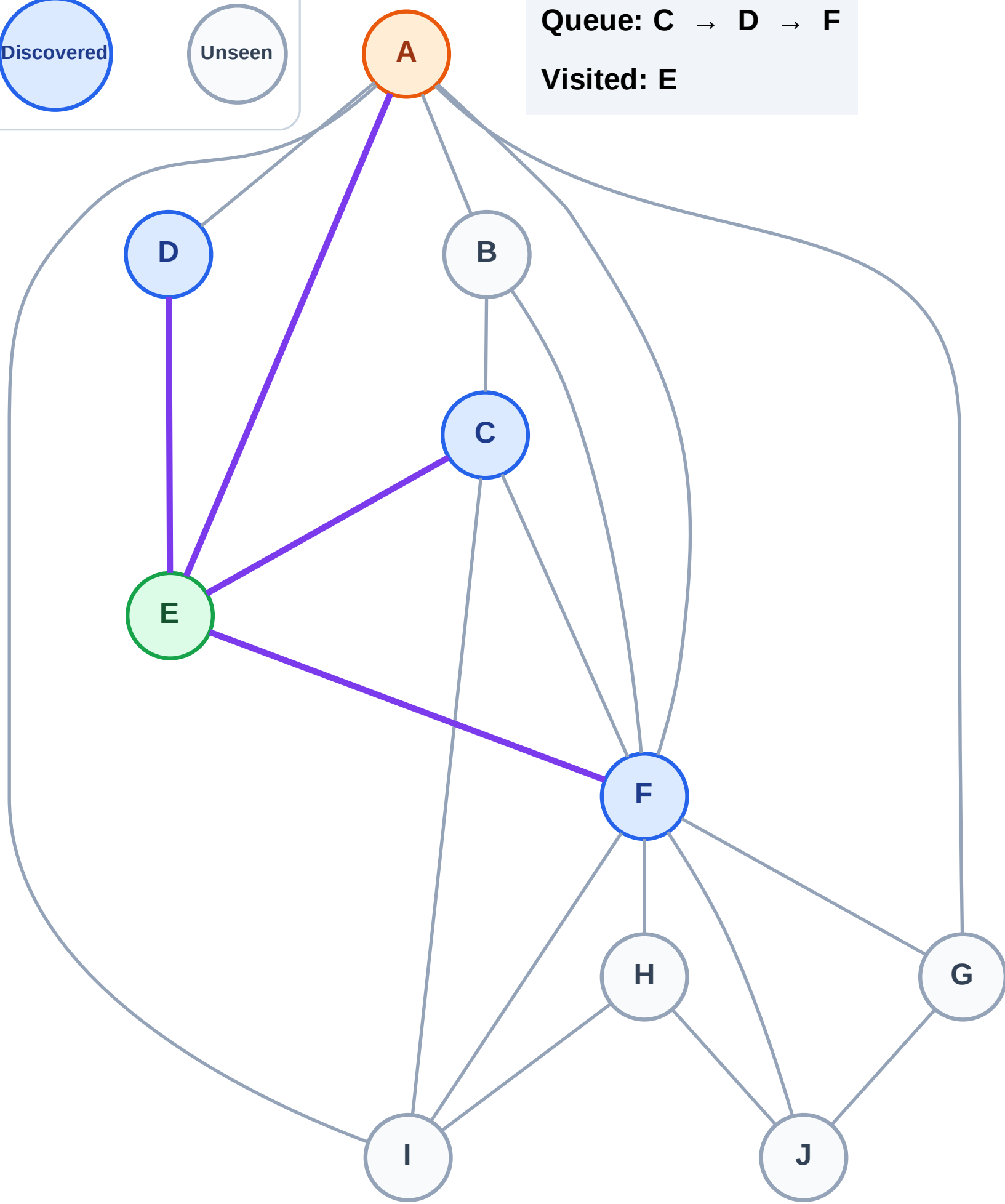
Step 4: Process node A

Legend



Queue: C → D → F

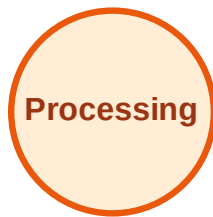
Visited: E



BFS Traversal

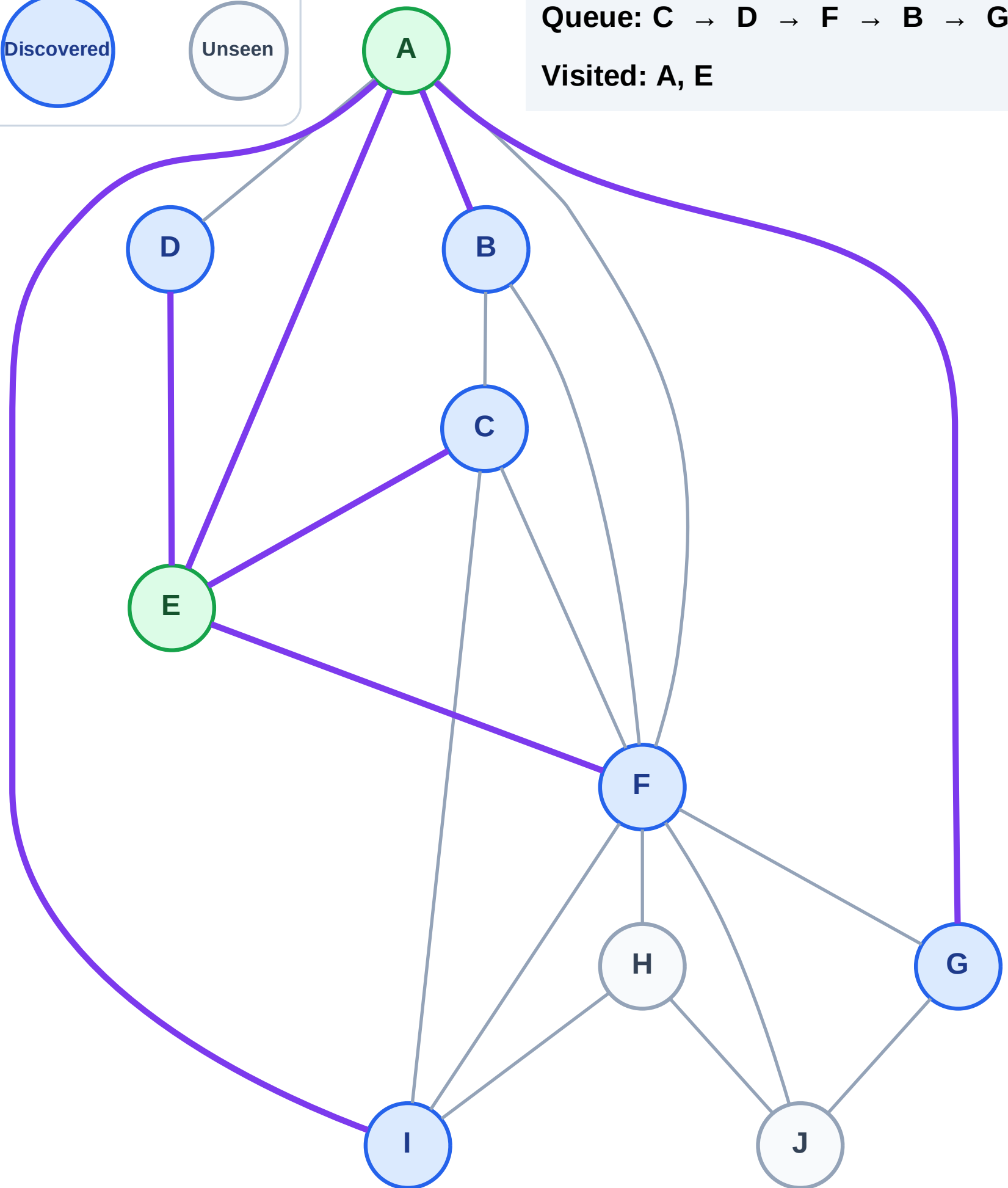
Step 5: Discover B, G, I from A

Legend



Queue: C → D → F → B → G → I

Visited: A, E



BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



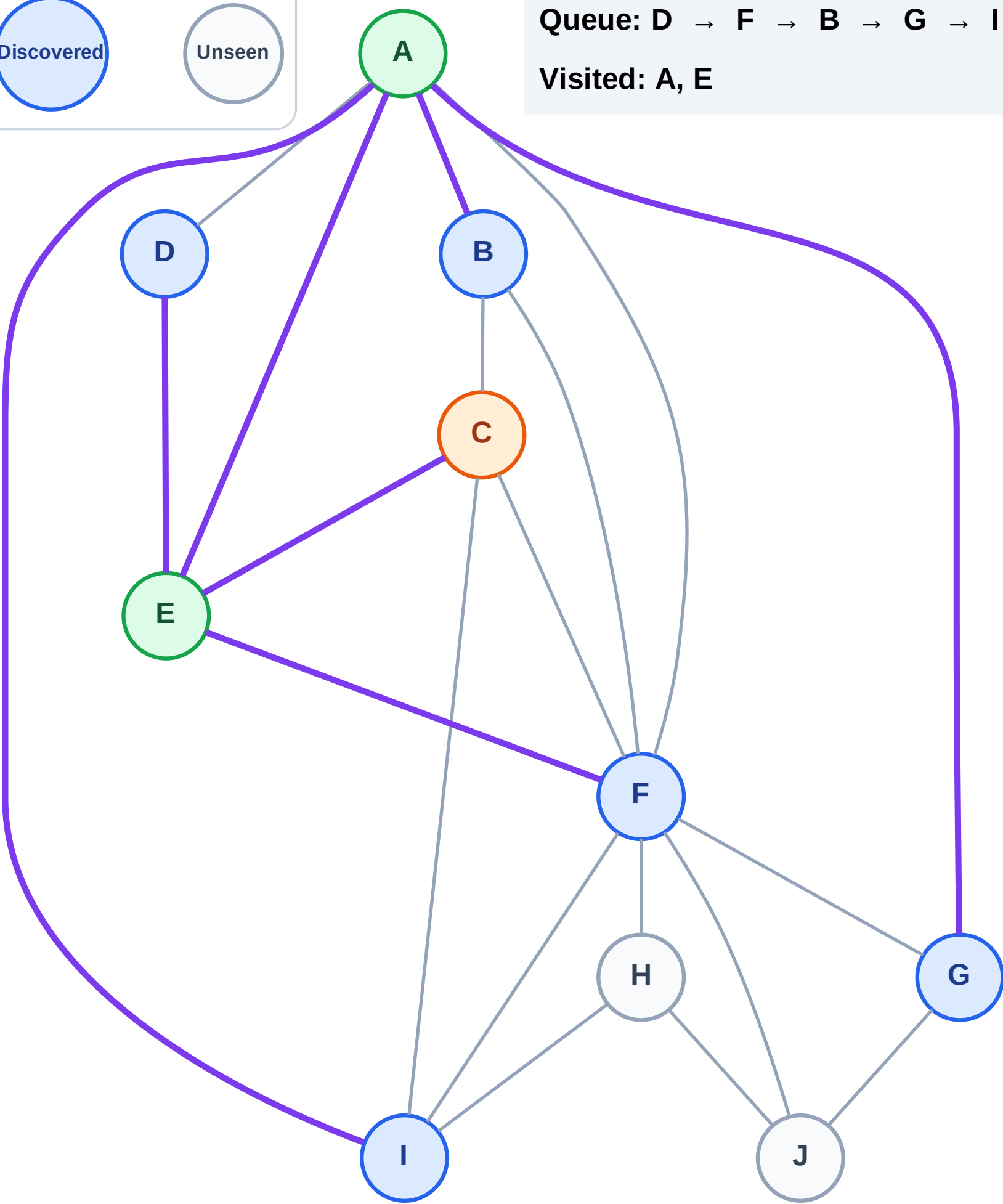
Discovered



Unseen

Queue: D → F → B → G → I

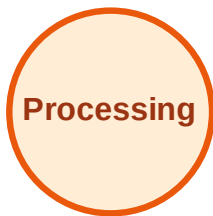
Visited: A, E



BFS Traversal

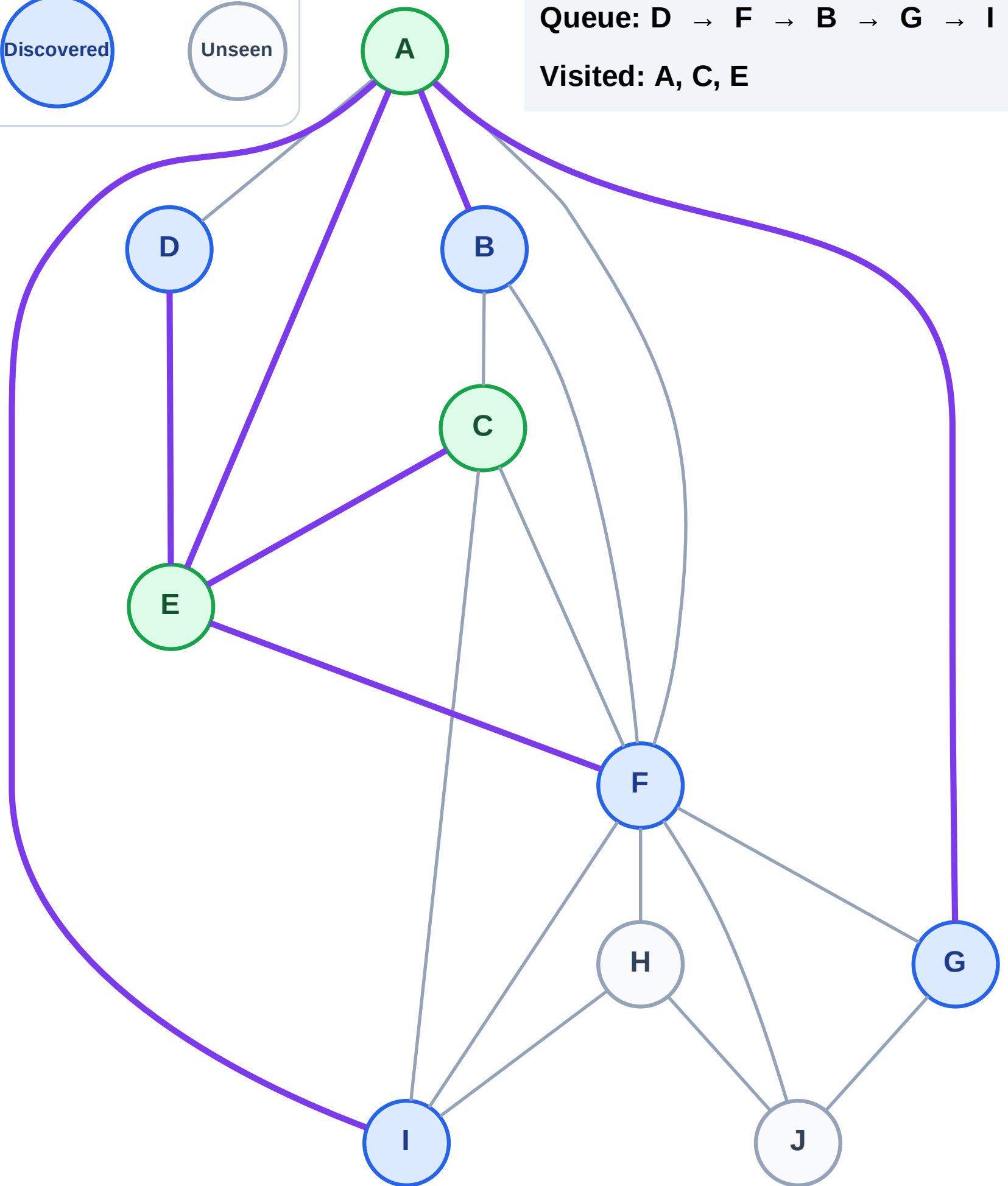
Step 7: No new neighbors from C

Legend



Queue: D → F → B → G → I

Visited: A, C, E



BFS Traversal

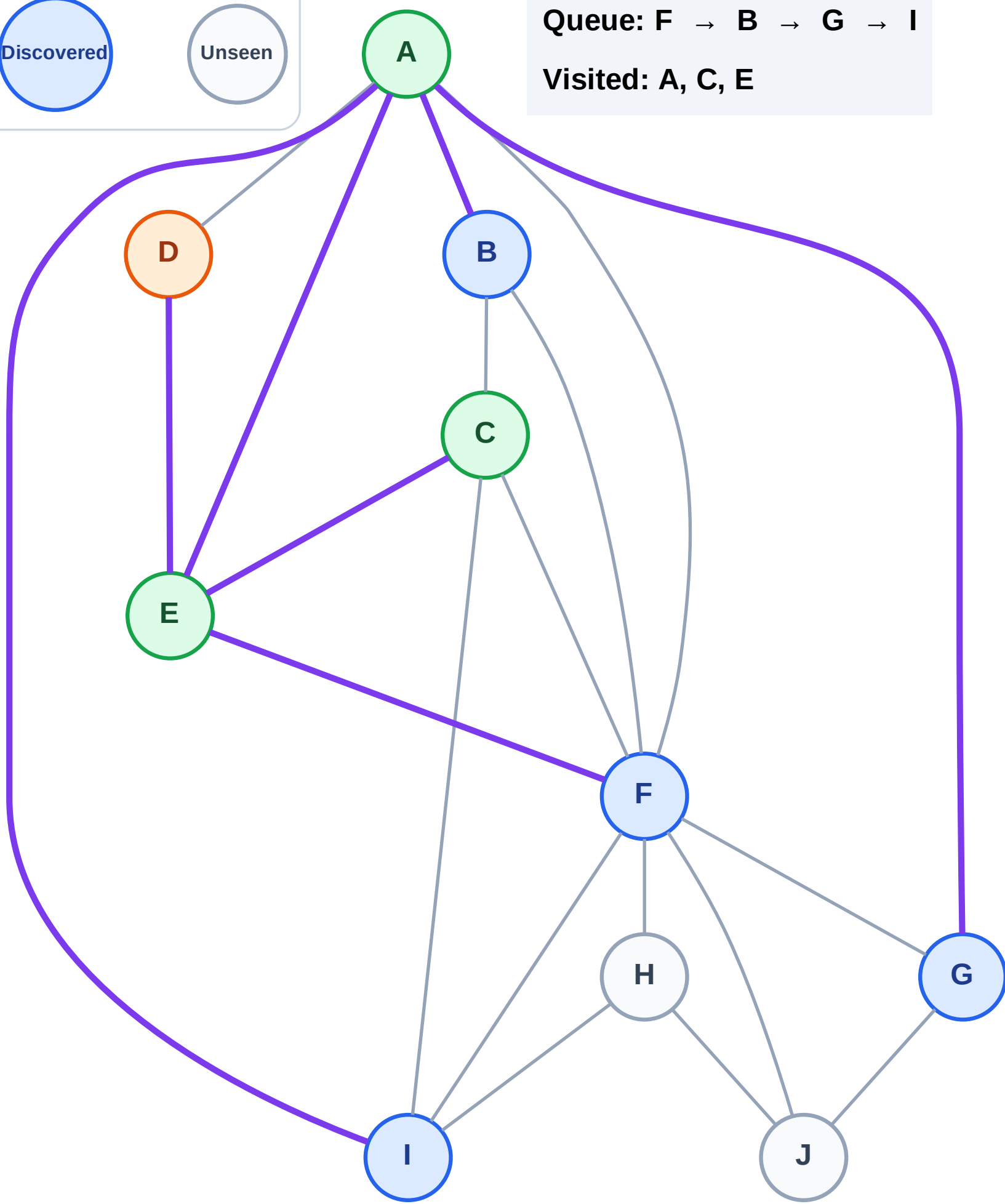
Step 8: Process node D

Legend



Queue: F → B → G → I

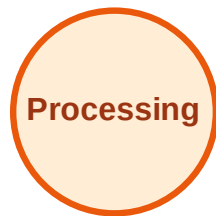
Visited: A, C, E



BFS Traversal

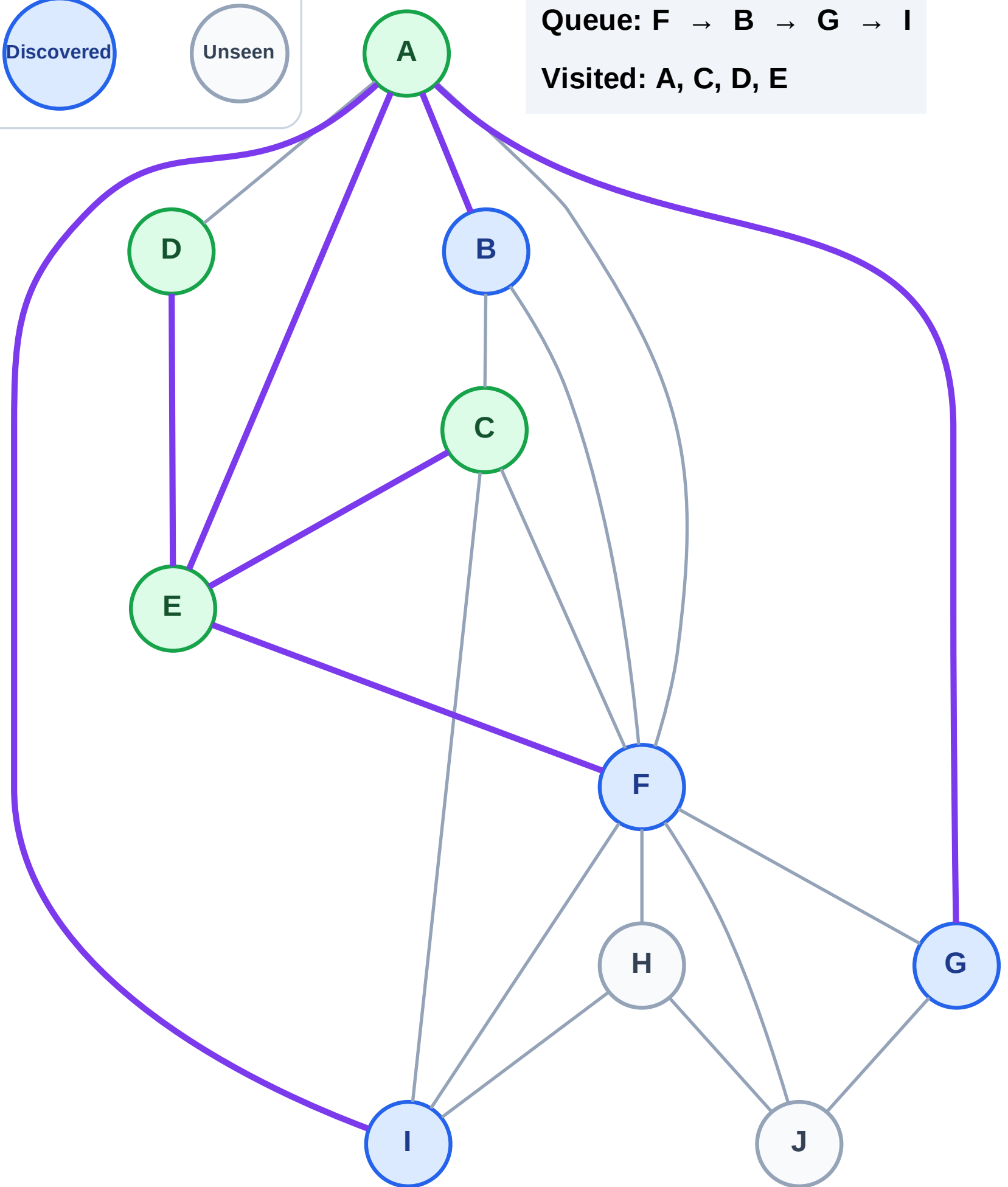
Step 9: No new neighbors from D

Legend



Queue: F → B → G → I

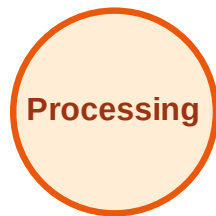
Visited: A, C, D, E



BFS Traversal

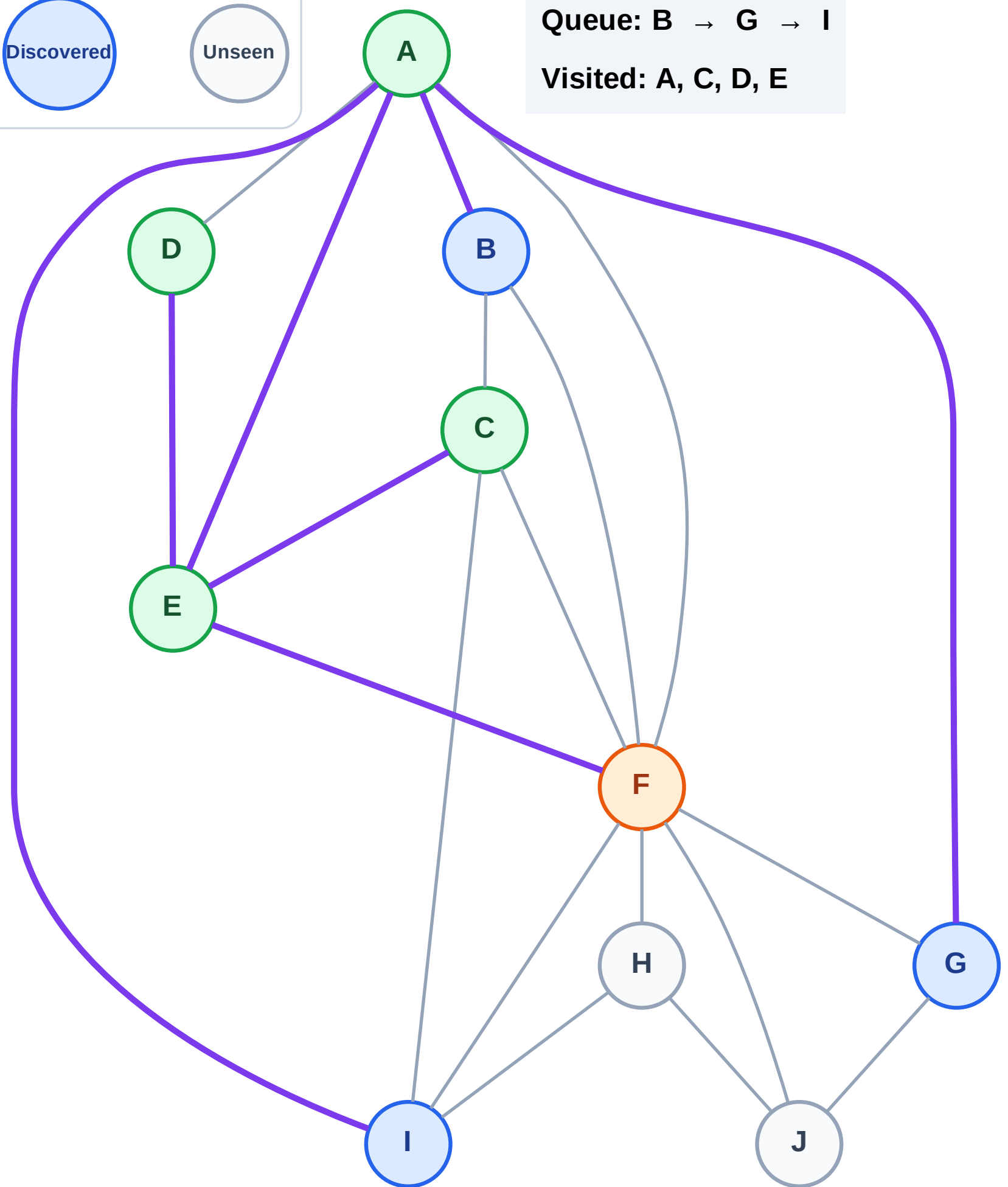
Step 10: Process node F

Legend



Queue: B → G → I

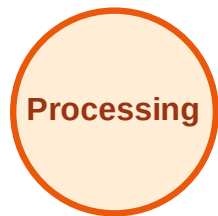
Visited: A, C, D, E



BFS Traversal

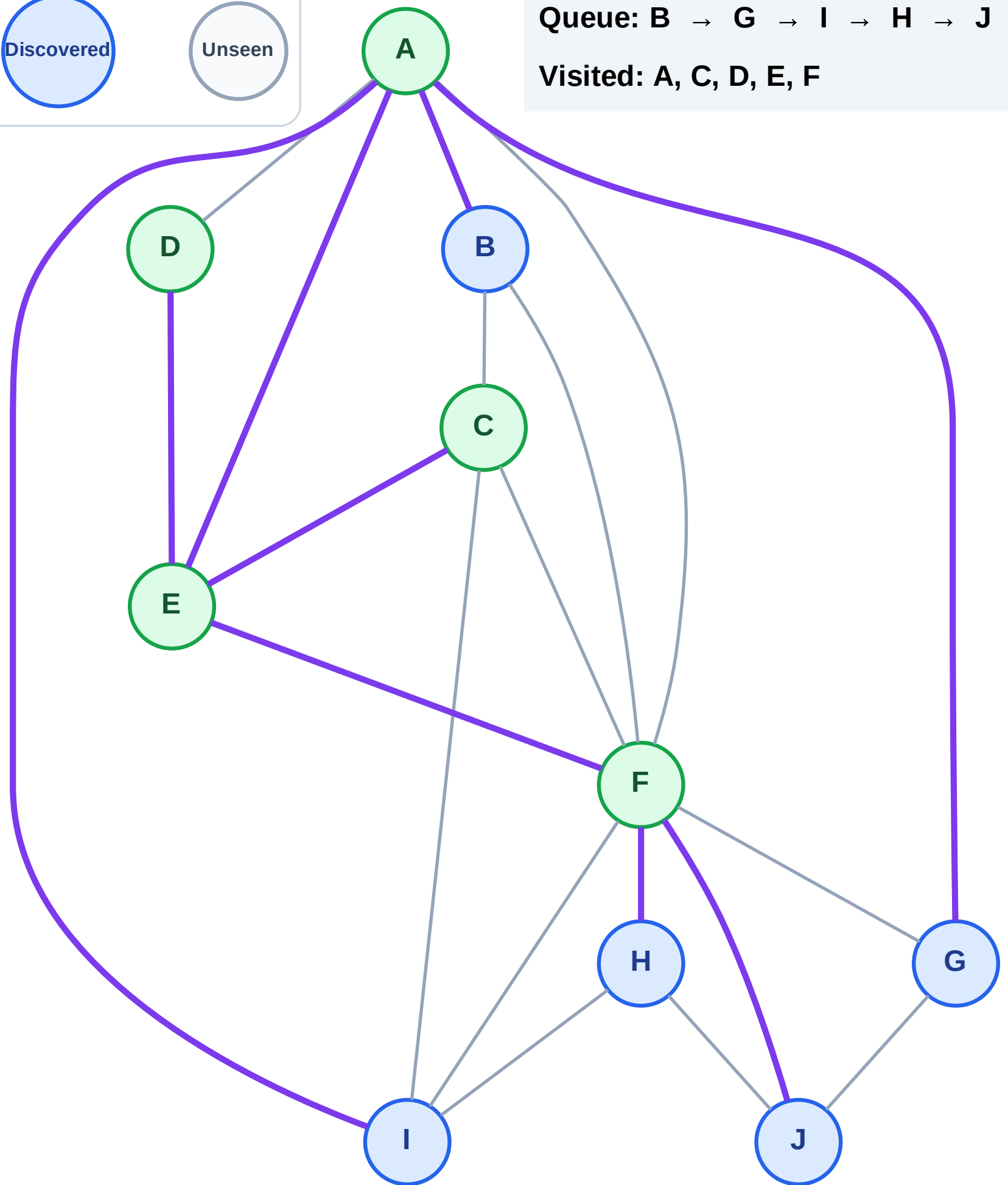
Step 11: Discover H, J from F

Legend



Queue: B → G → I → H → J

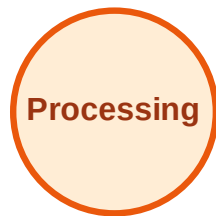
Visited: A, C, D, E, F



BFS Traversal

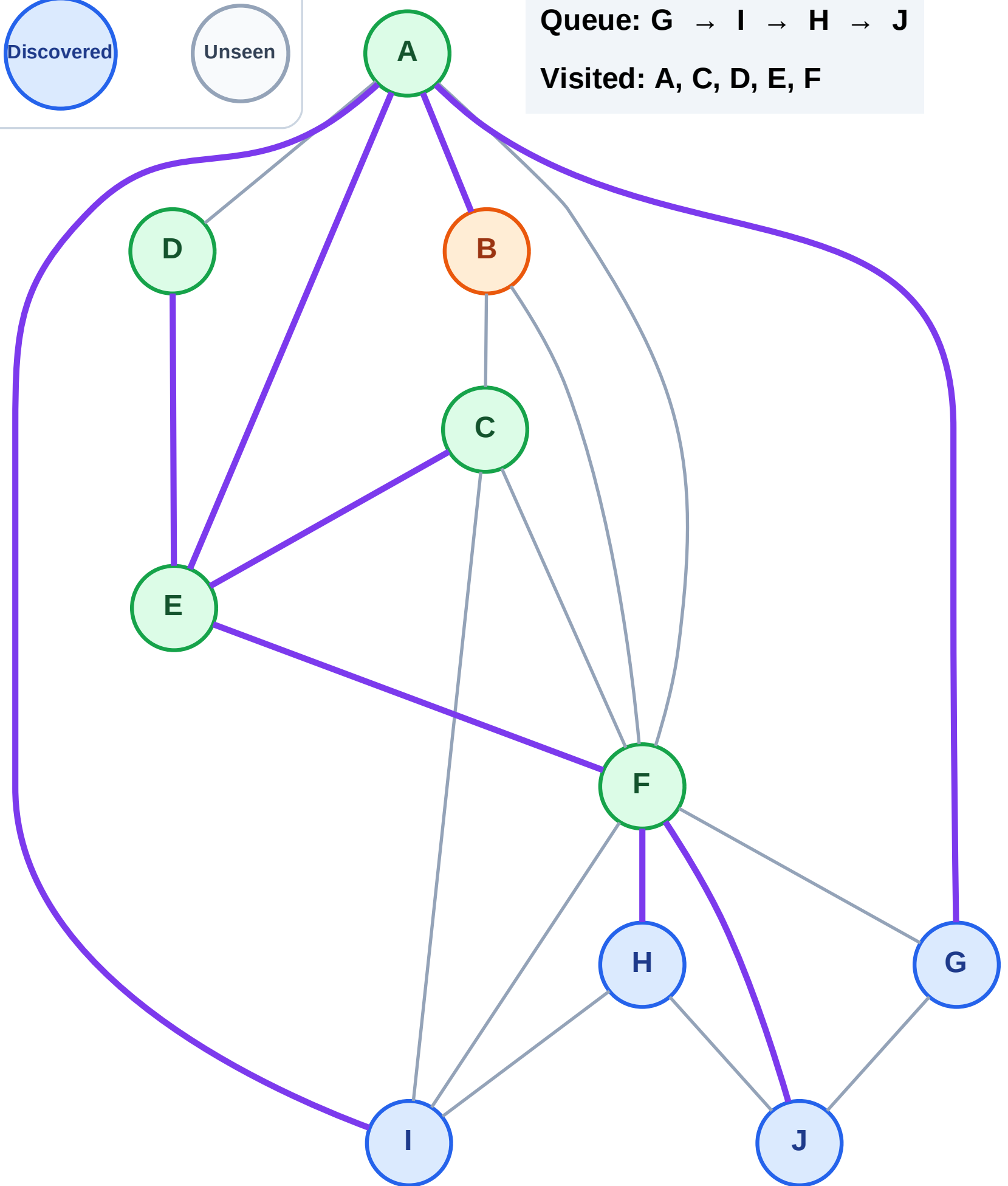
Step 12: Process node B

Legend



Queue: G → I → H → J

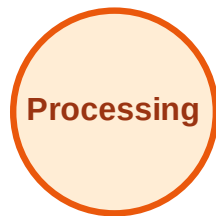
Visited: A, C, D, E, F



BFS Traversal

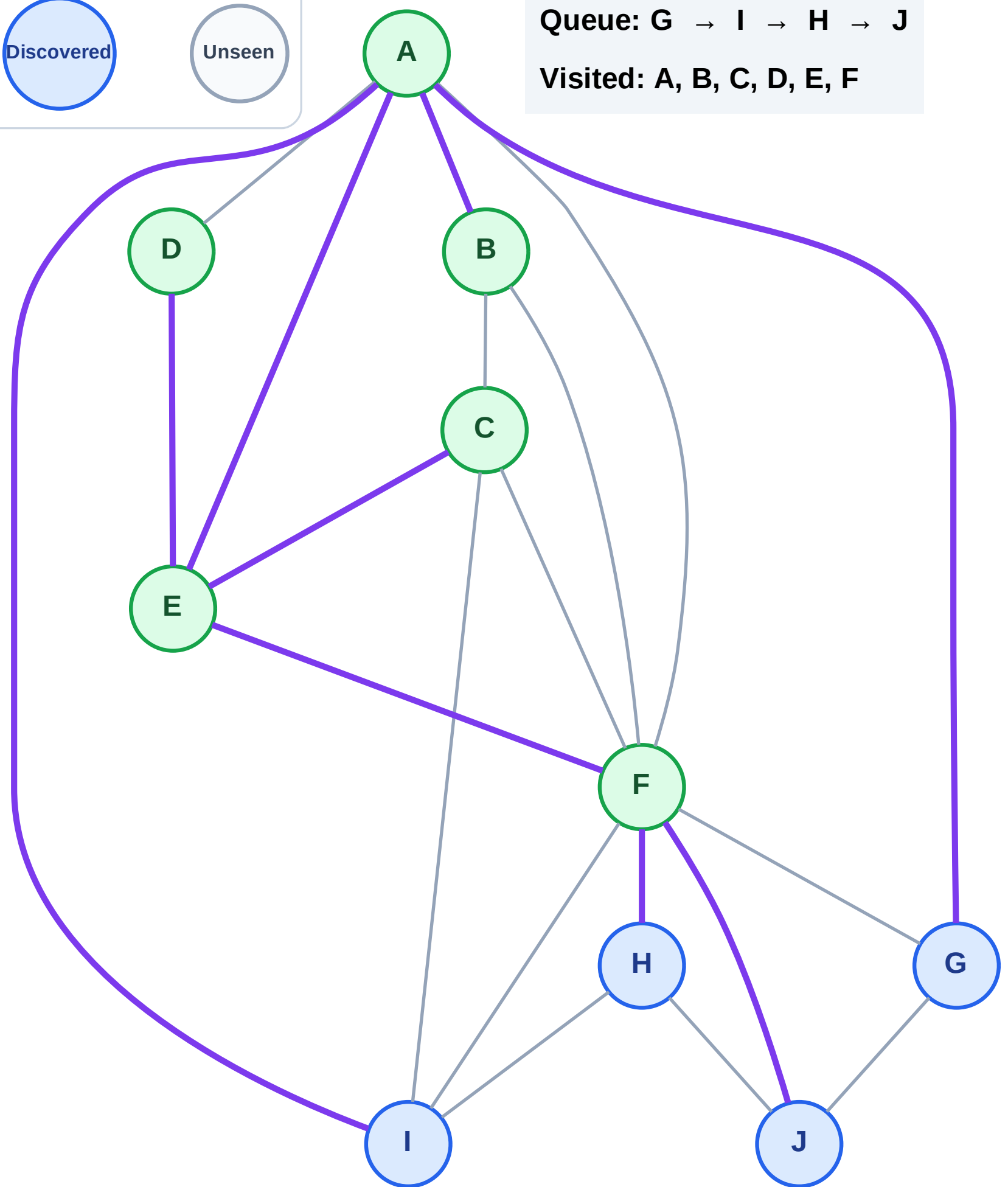
Step 13: No new neighbors from B

Legend



Queue: G → I → H → J

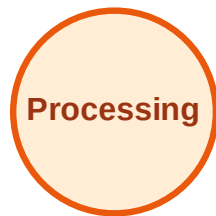
Visited: A, B, C, D, E, F



BFS Traversal

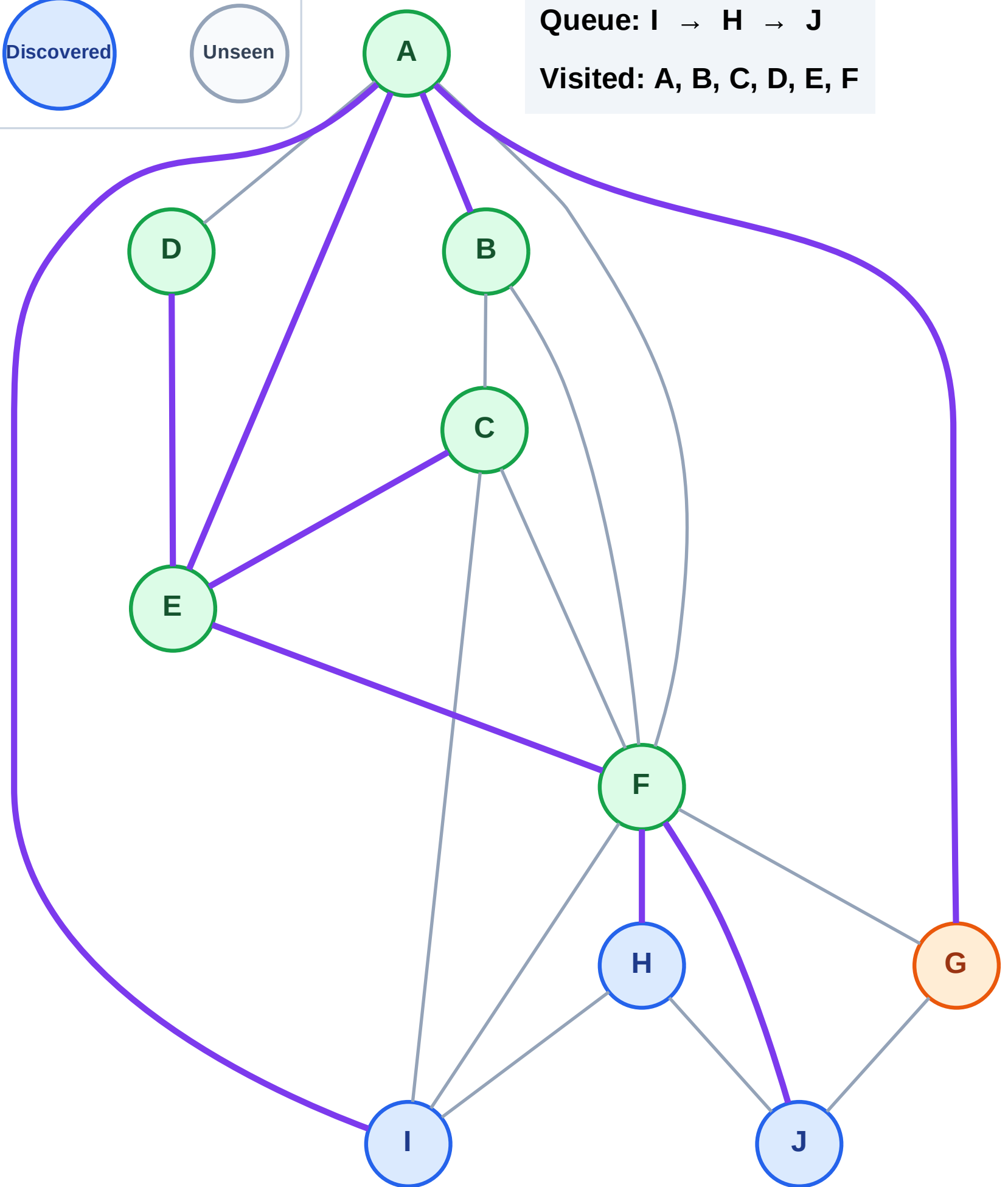
Step 14: Process node G

Legend



Queue: I → H → J

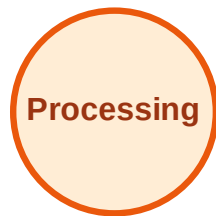
Visited: A, B, C, D, E, F



BFS Traversal

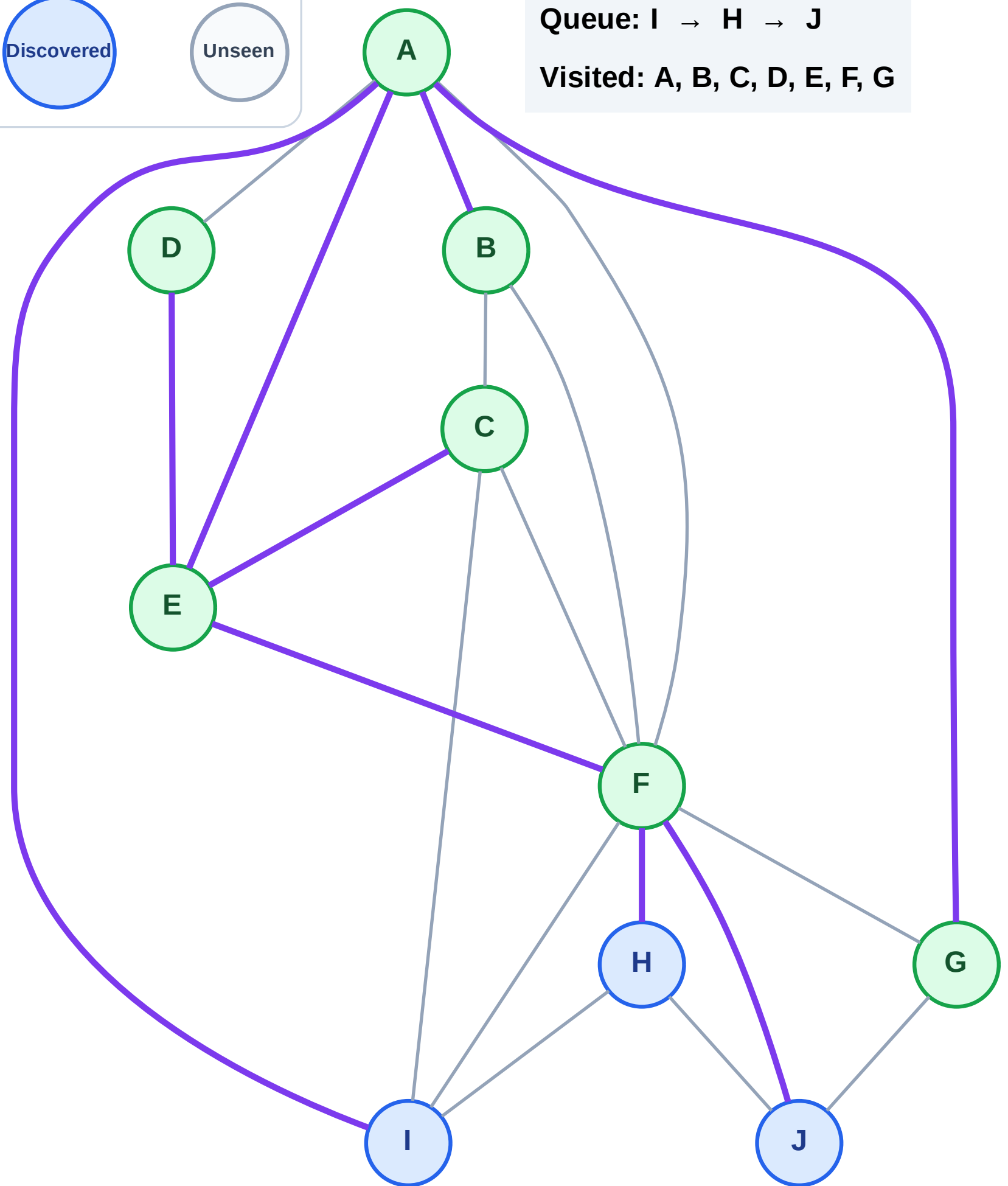
Step 15: No new neighbors from G

Legend



Queue: I → H → J

Visited: A, B, C, D, E, F, G



BFS Traversal

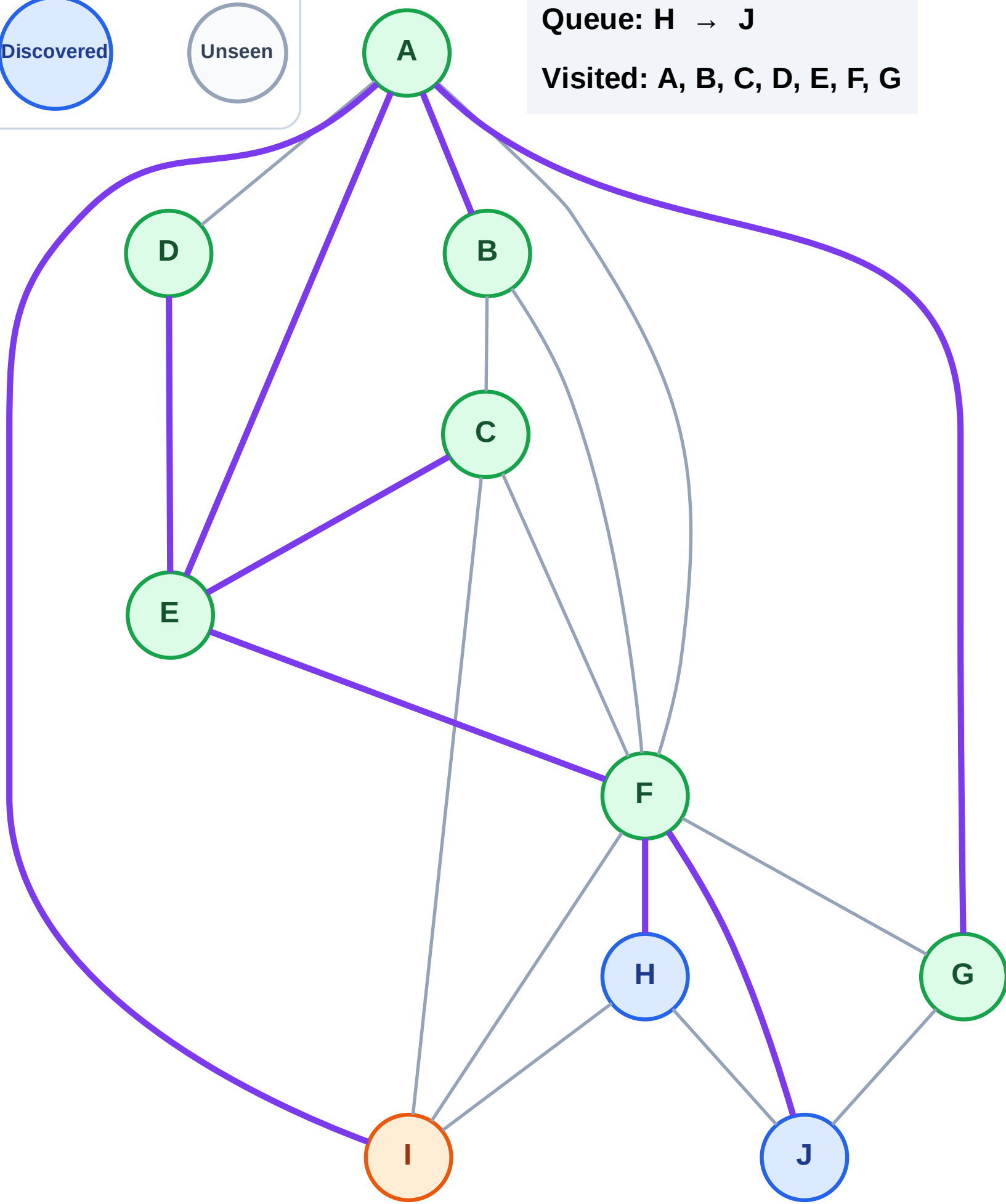
Step 16: Process node I

Legend



Queue: H → J

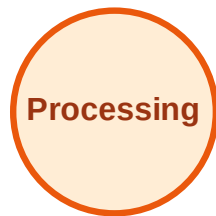
Visited: A, B, C, D, E, F, G



BFS Traversal

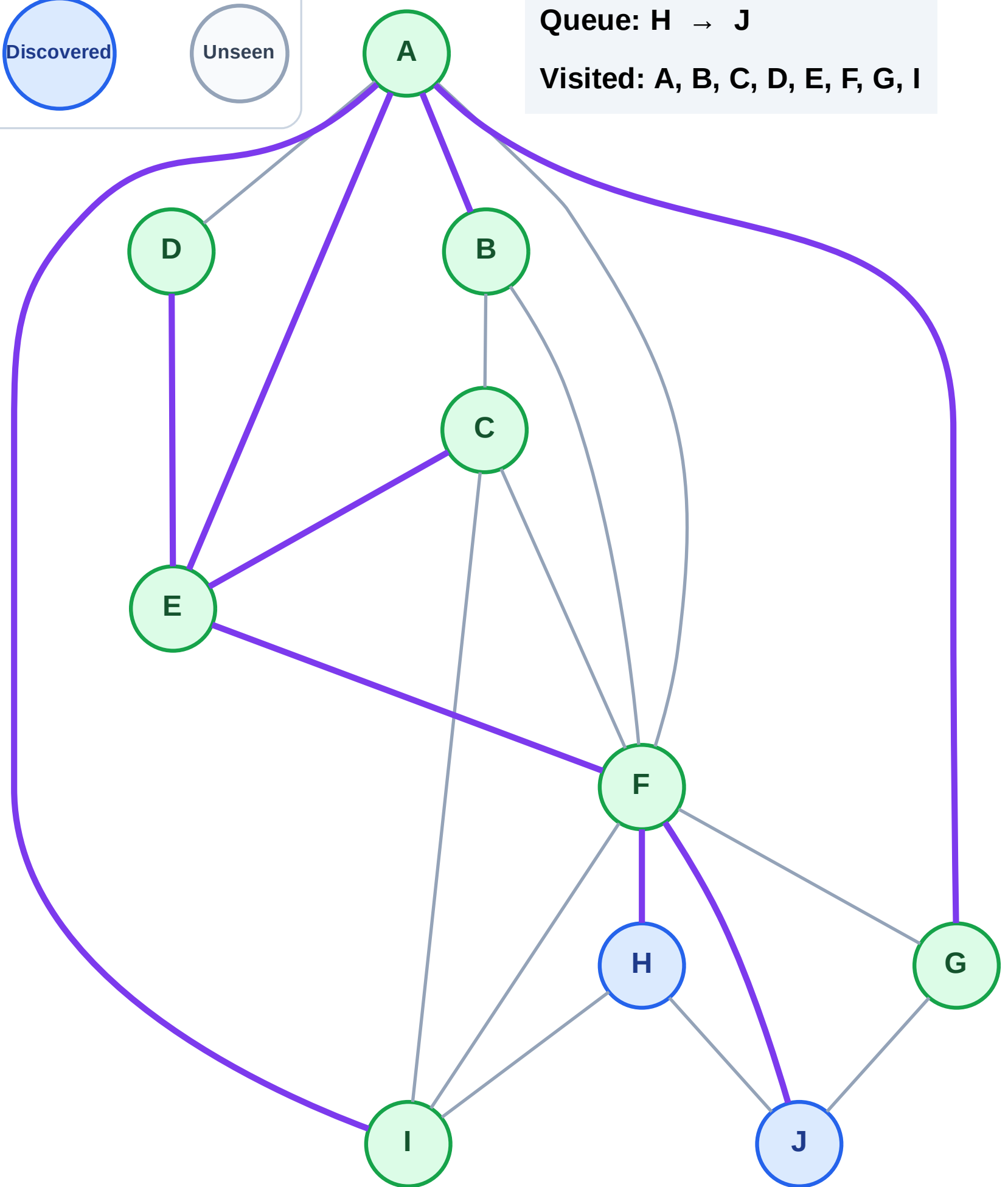
Step 17: No new neighbors from I

Legend



Queue: H → J

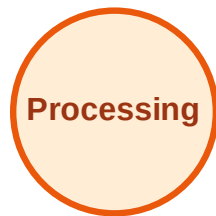
Visited: A, B, C, D, E, F, G, I



BFS Traversal

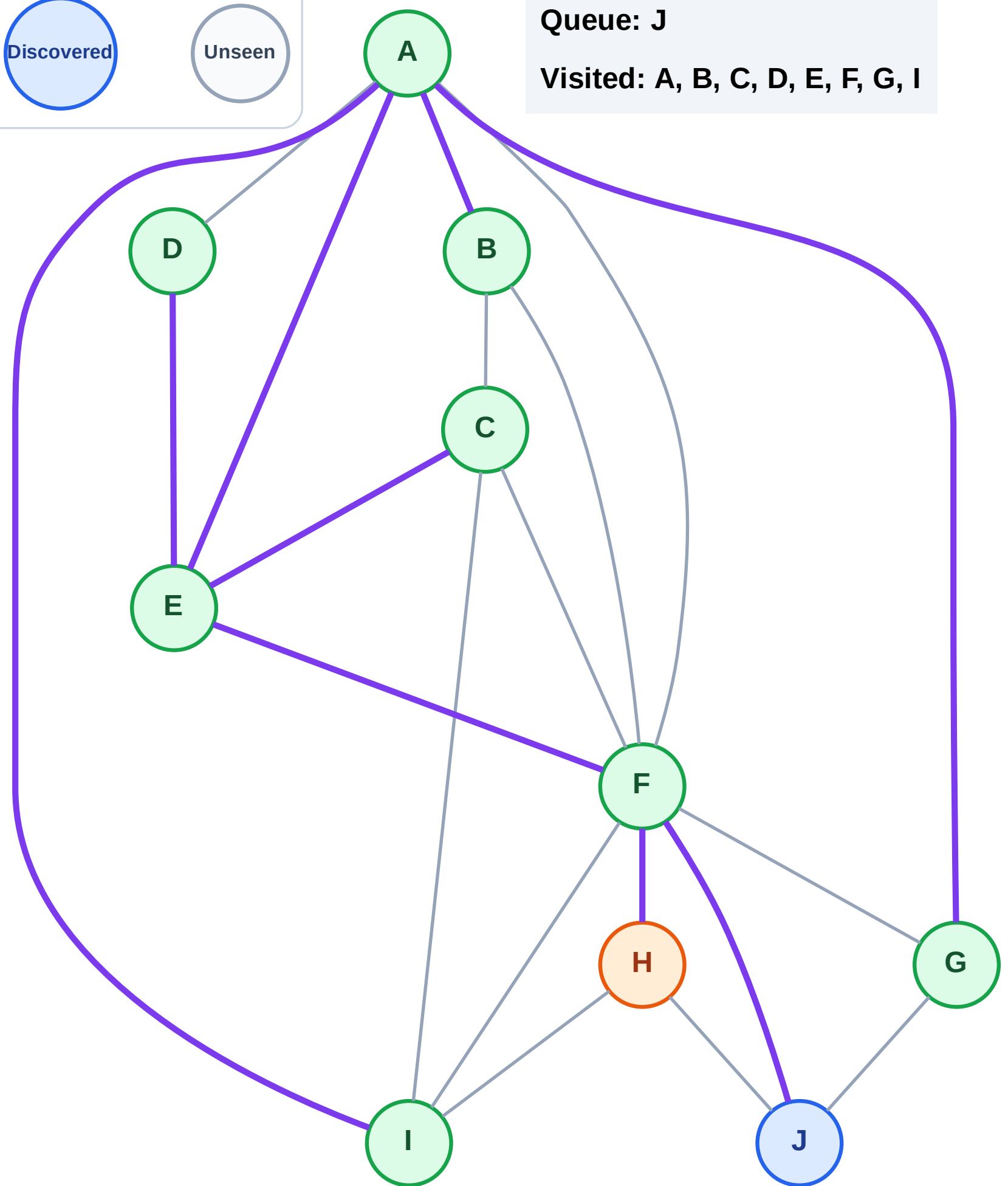
Step 18: Process node H

Legend



Queue: J

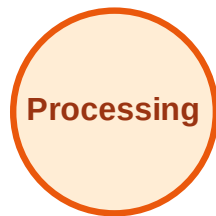
Visited: A, B, C, D, E, F, G, I



BFS Traversal

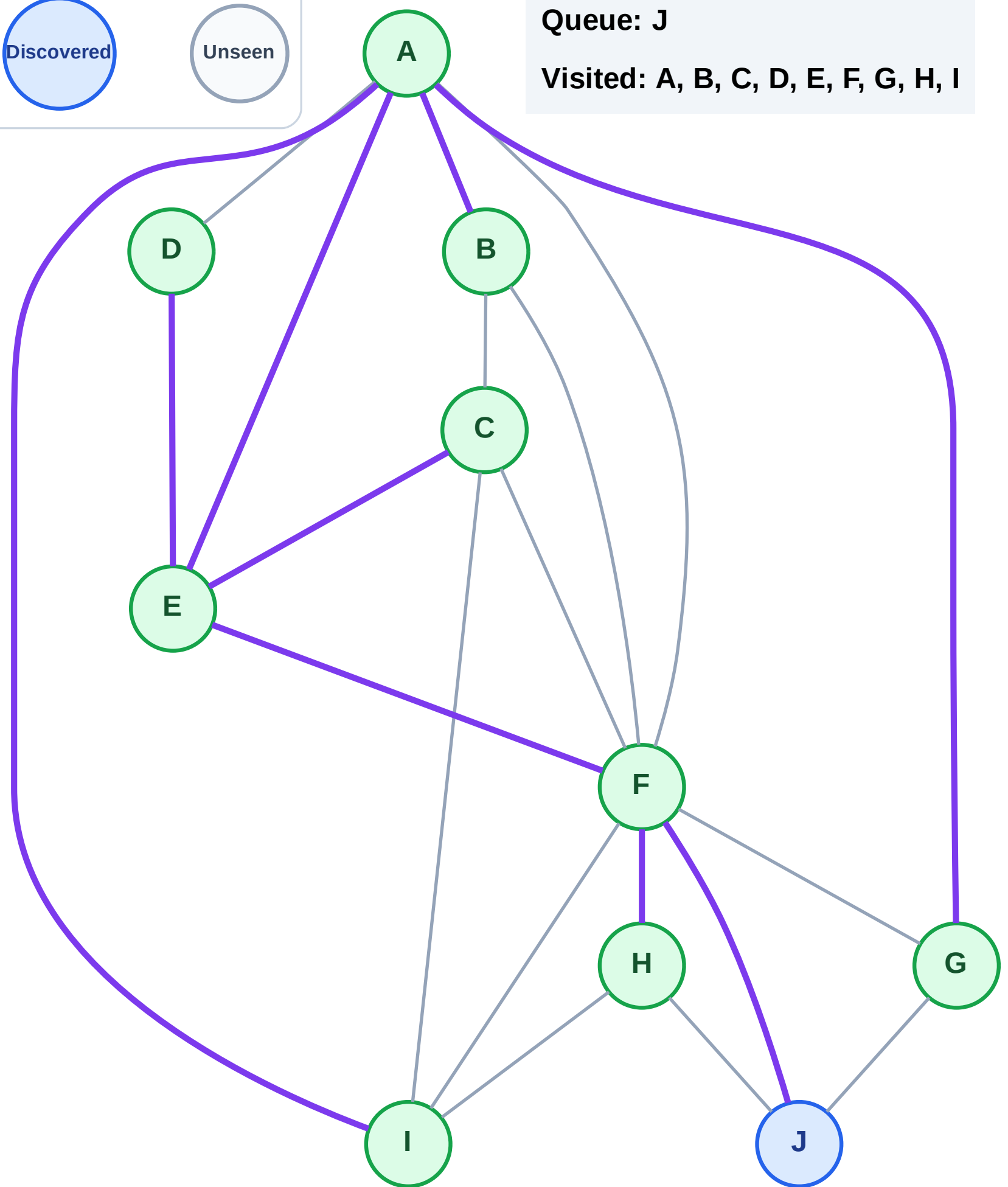
Step 19: No new neighbors from H

Legend



Queue: J

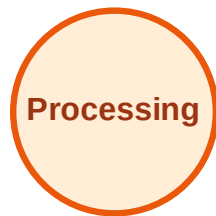
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

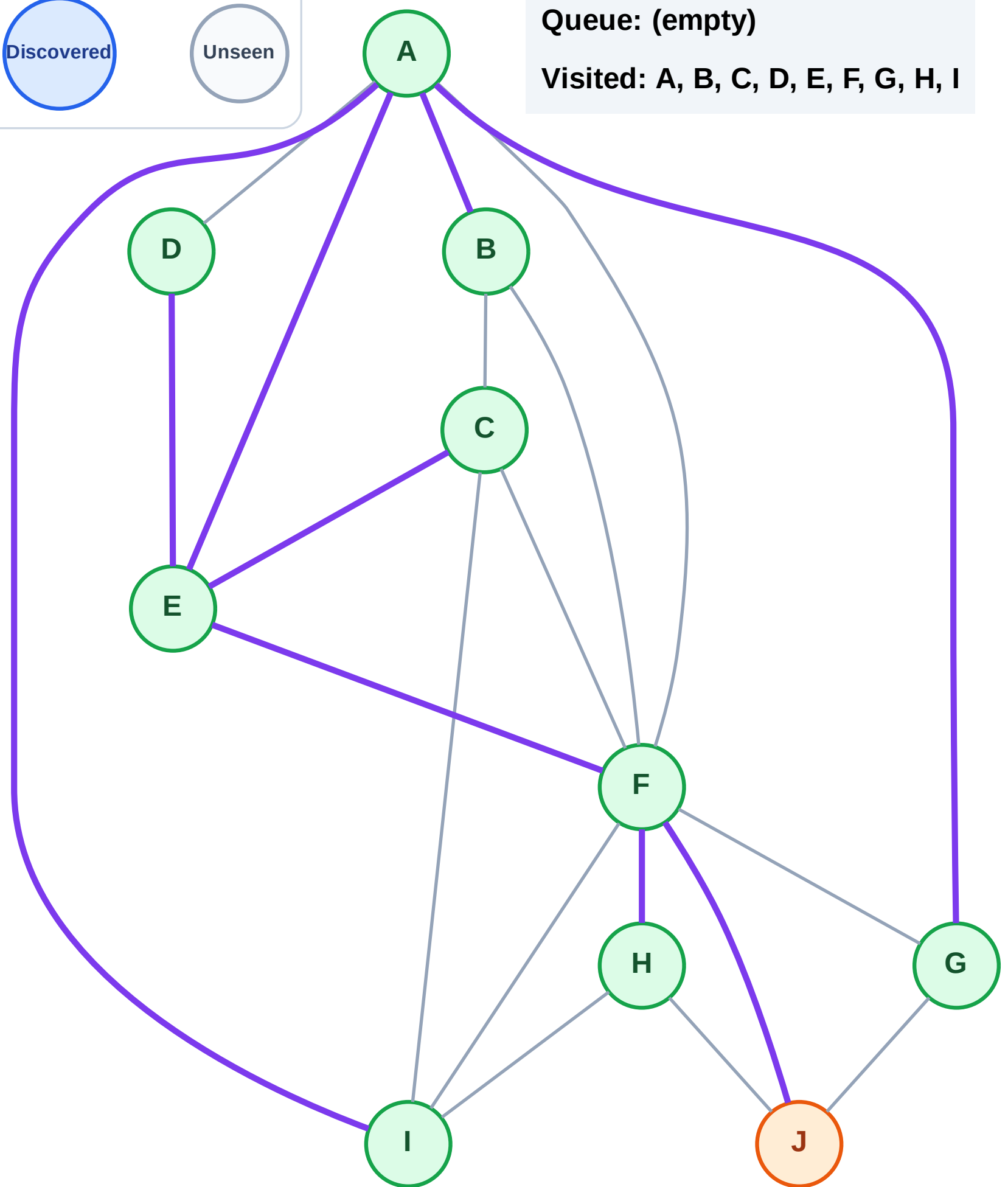
Step 20: Process node J

Legend



Queue: (empty)

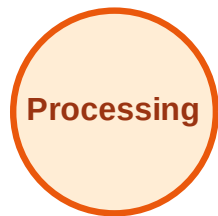
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

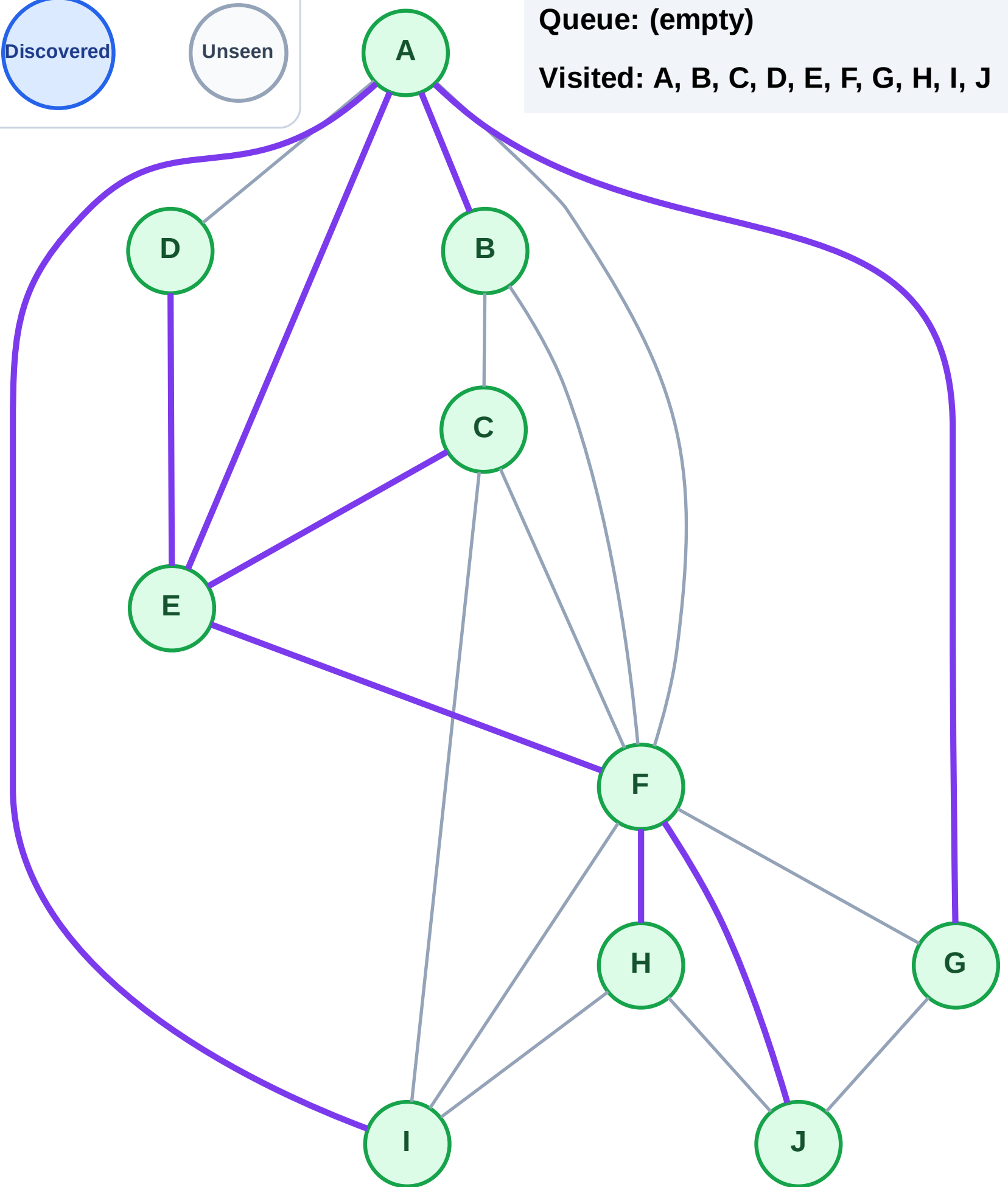
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

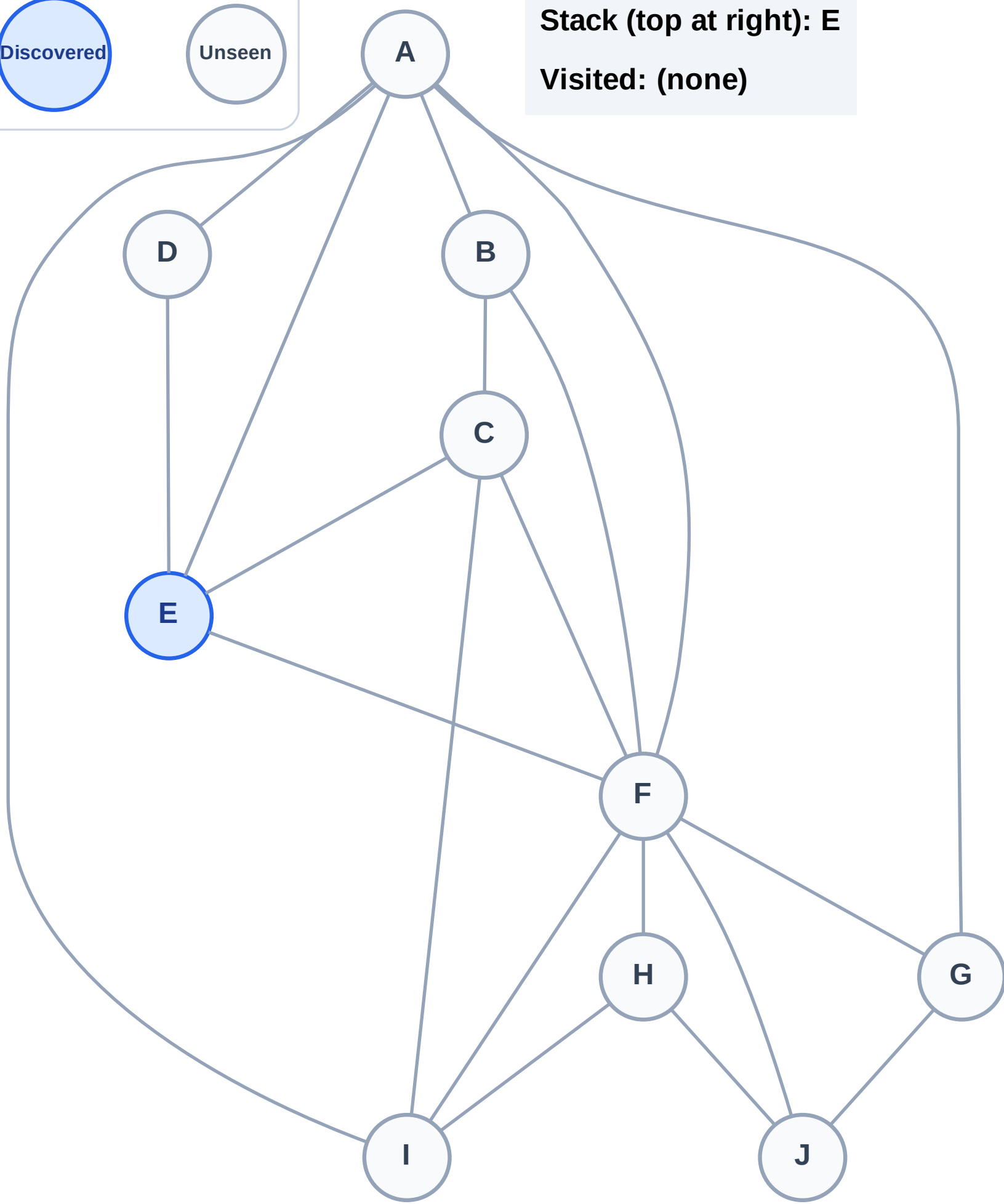
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

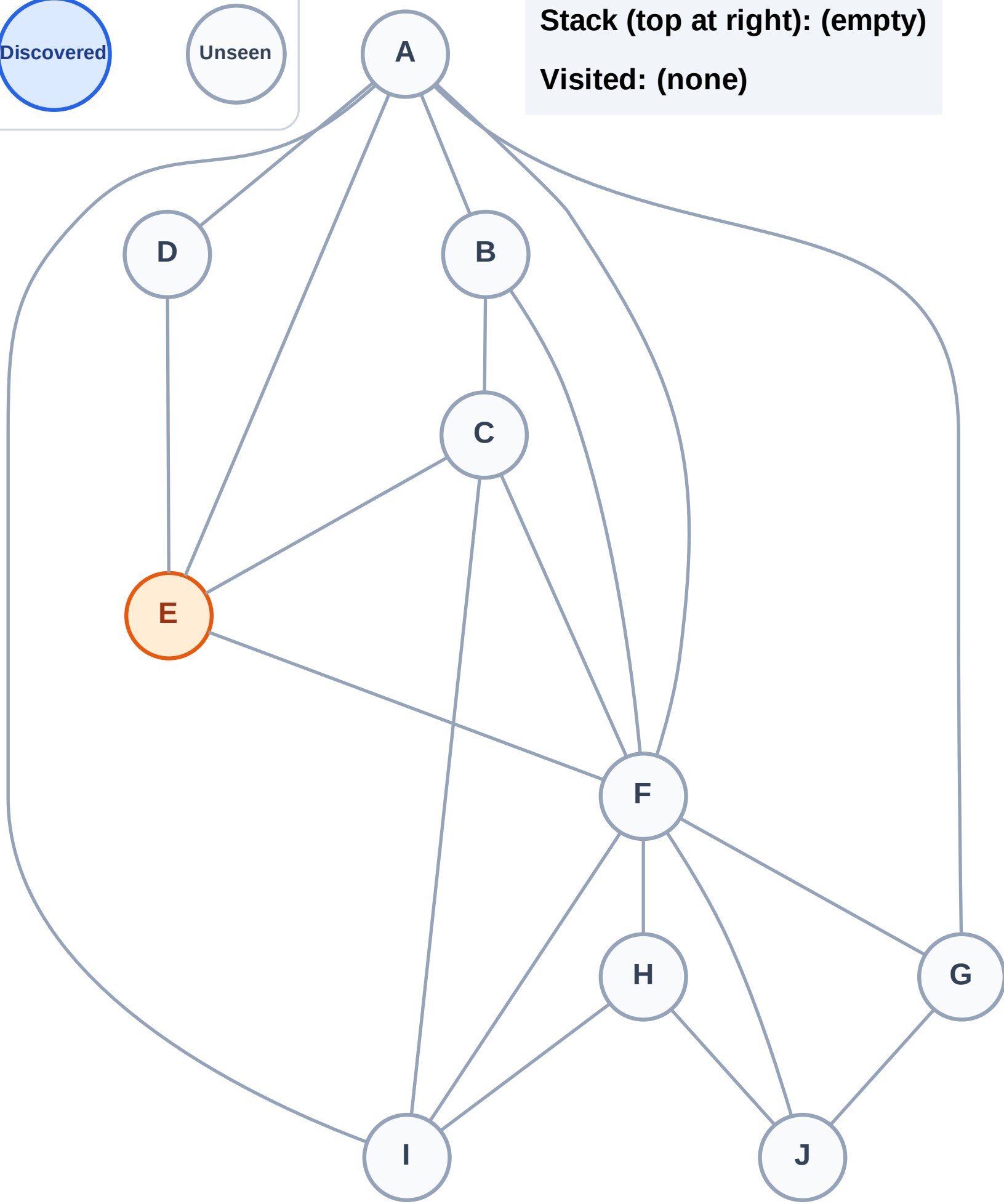
Step 2: Process node E

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

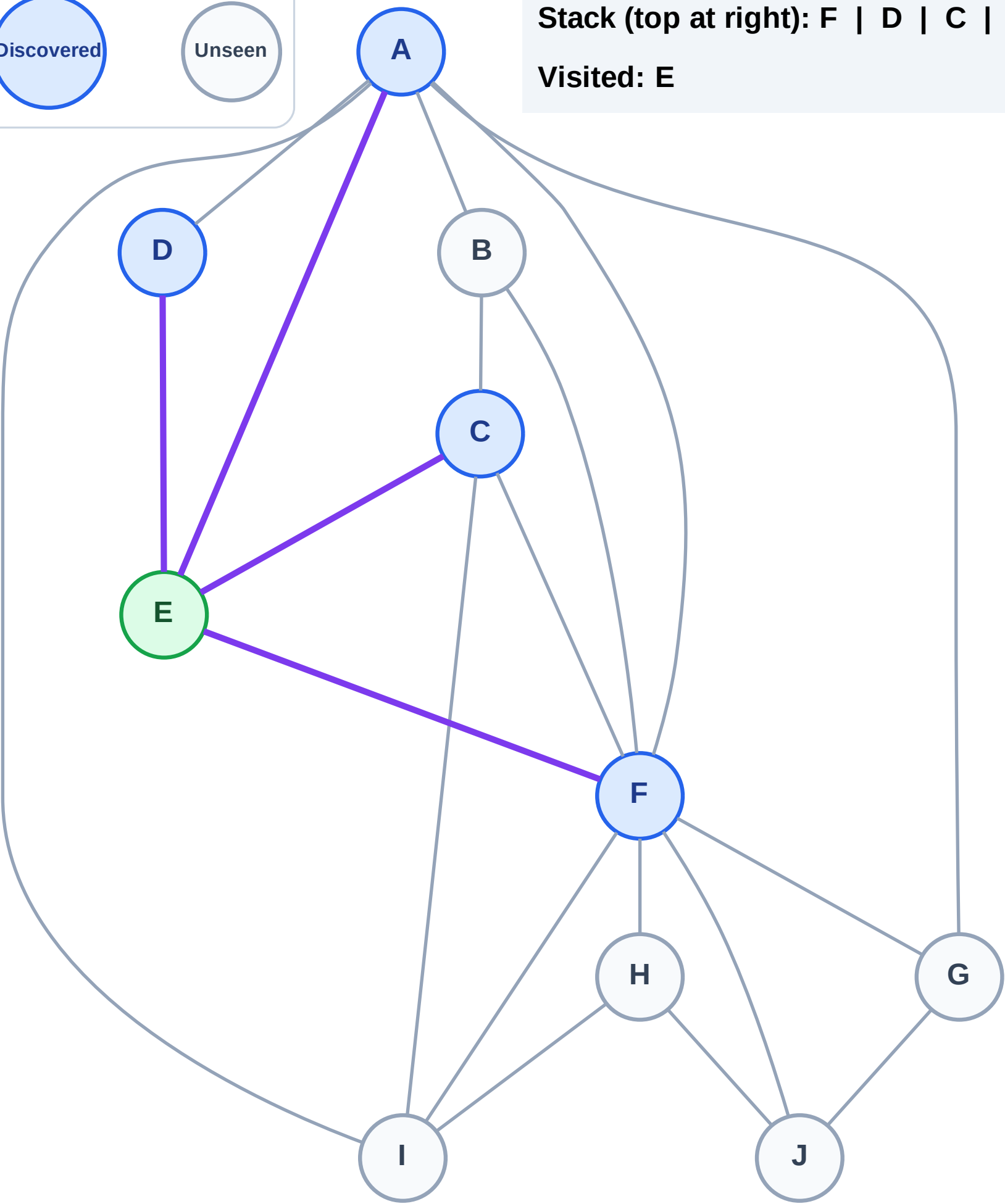
Step 3: Discover F, D, C, A from E

Legend



Stack (top at right): F | D | C | A

Visited: E



DFS Traversal

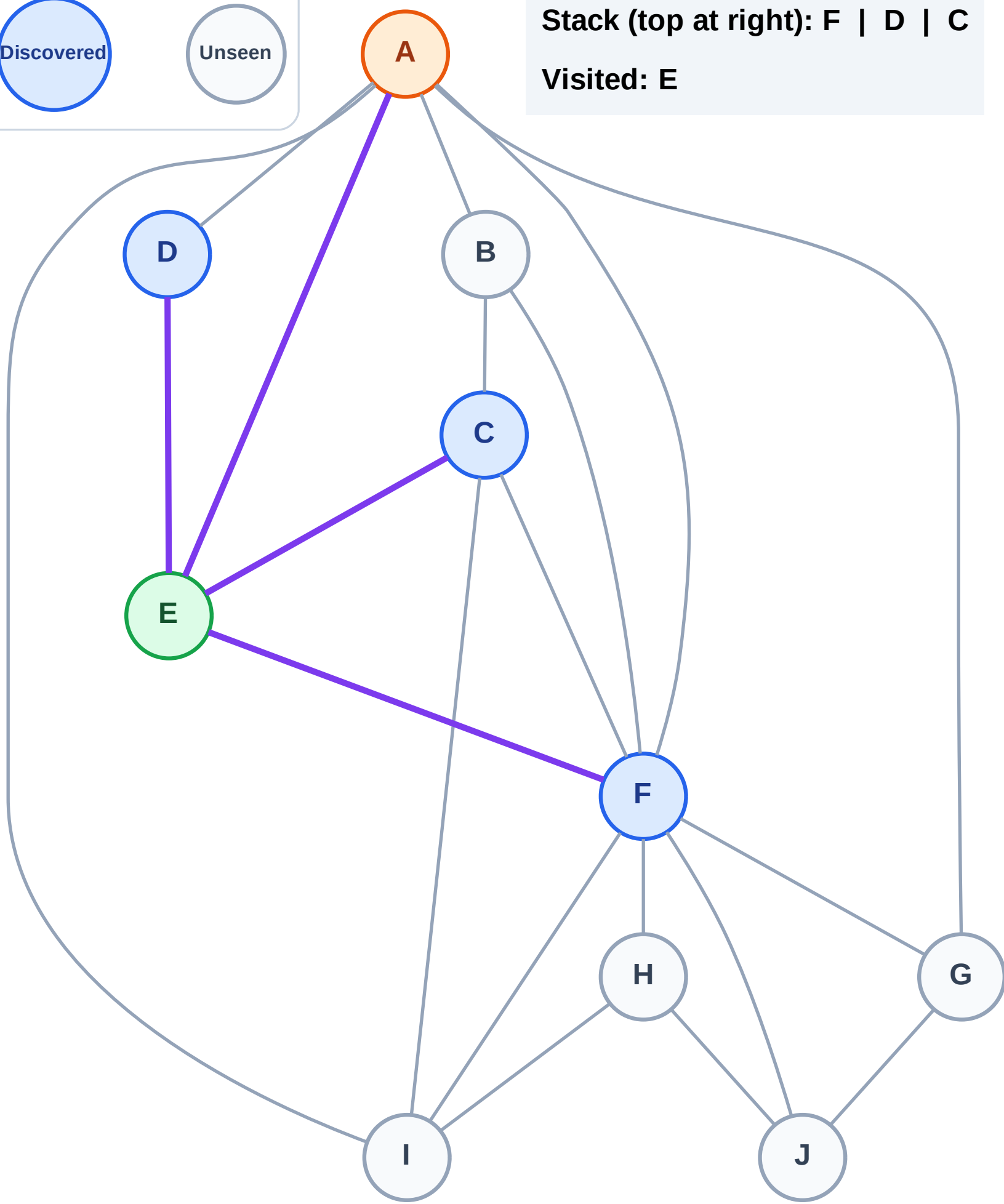
Step 4: Process node A

Legend



Stack (top at right): F | D | C

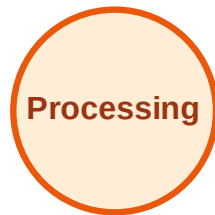
Visited: E



DFS Traversal

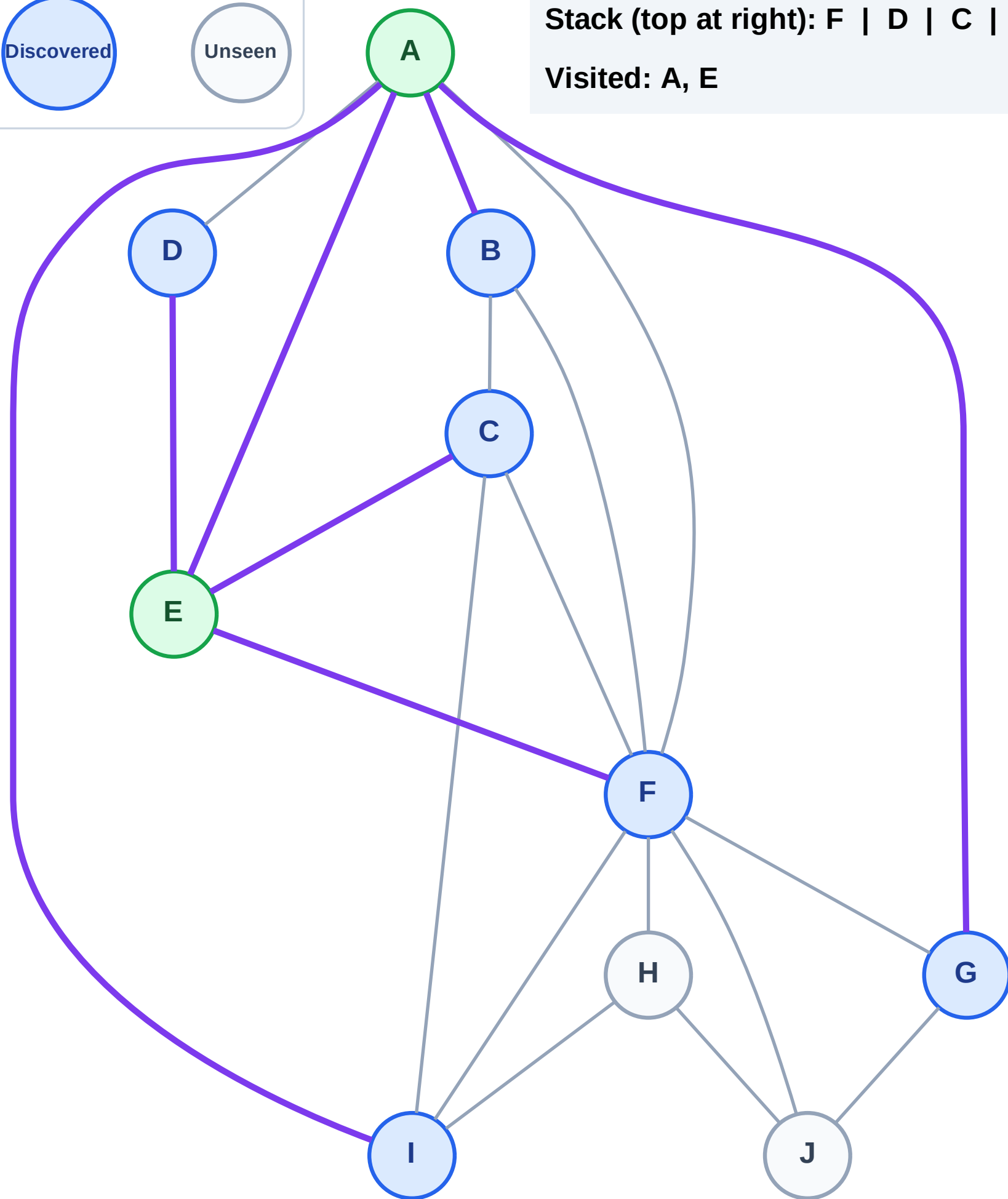
Step 5: Discover I, G, B from A

Legend



Stack (top at right): F | D | C | I | G | B

Visited: A, E



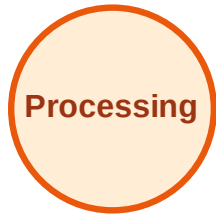
DFS Traversal

Step 6: Process node B

Legend



Complete



Processing



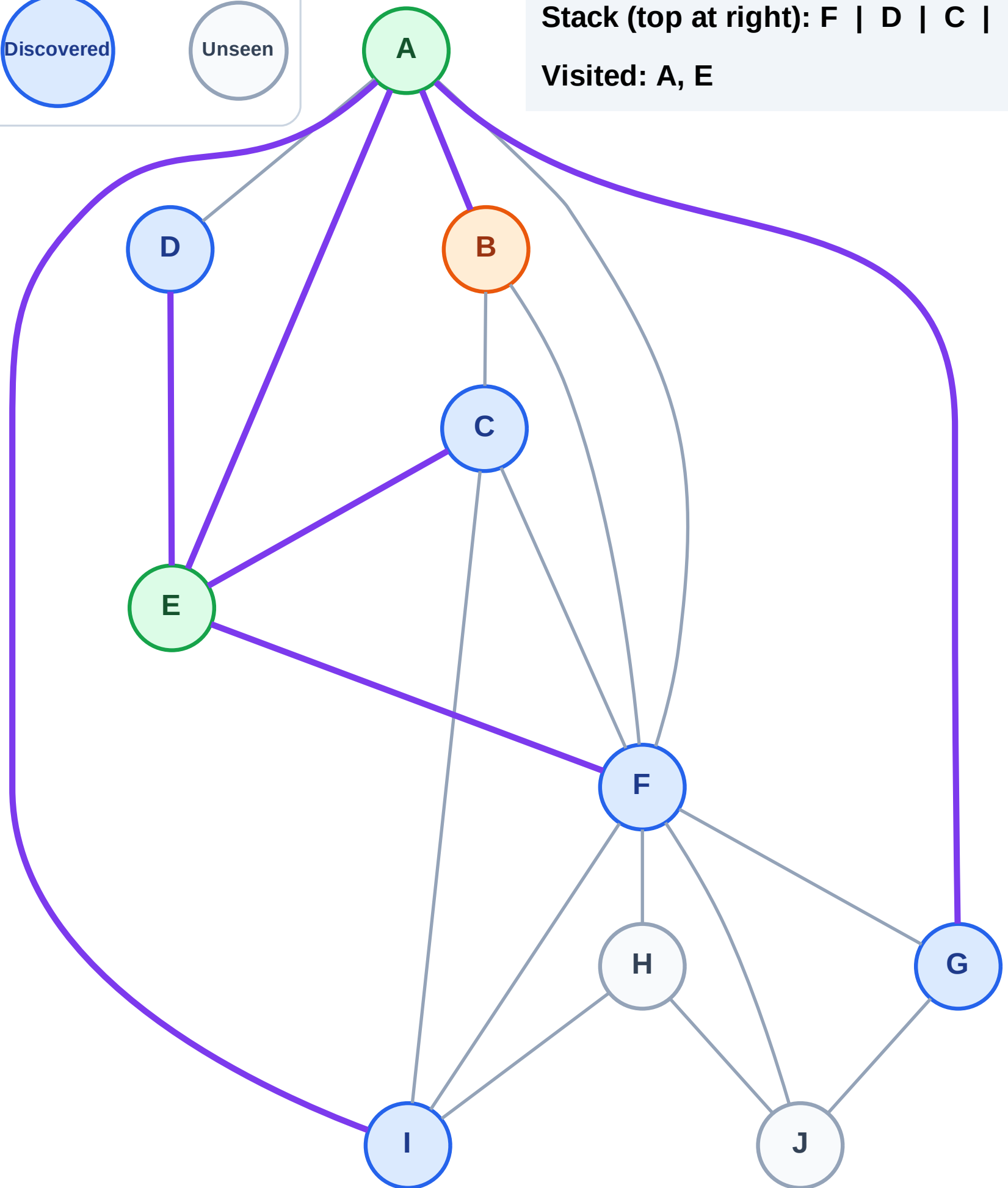
Discovered



Unseen

Stack (top at right): F | D | C | I | G

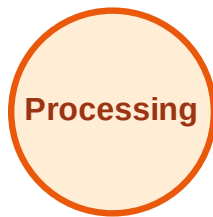
Visited: A, E



DFS Traversal

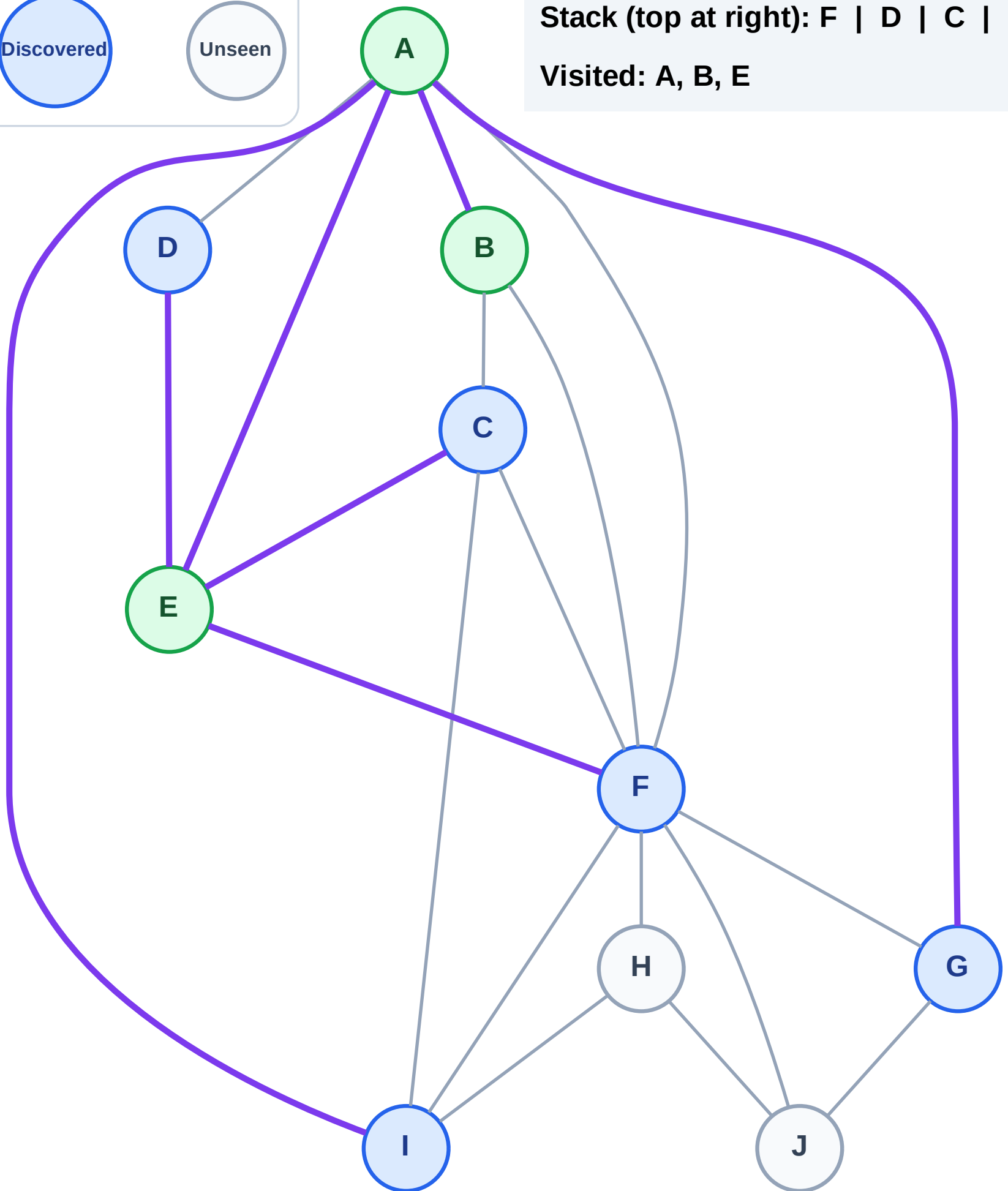
Step 7: No new neighbors from B

Legend



Stack (top at right): F | D | C | I | G

Visited: A, B, E



DFS Traversal

Step 8: Process node G

Legend



Complete



Processing



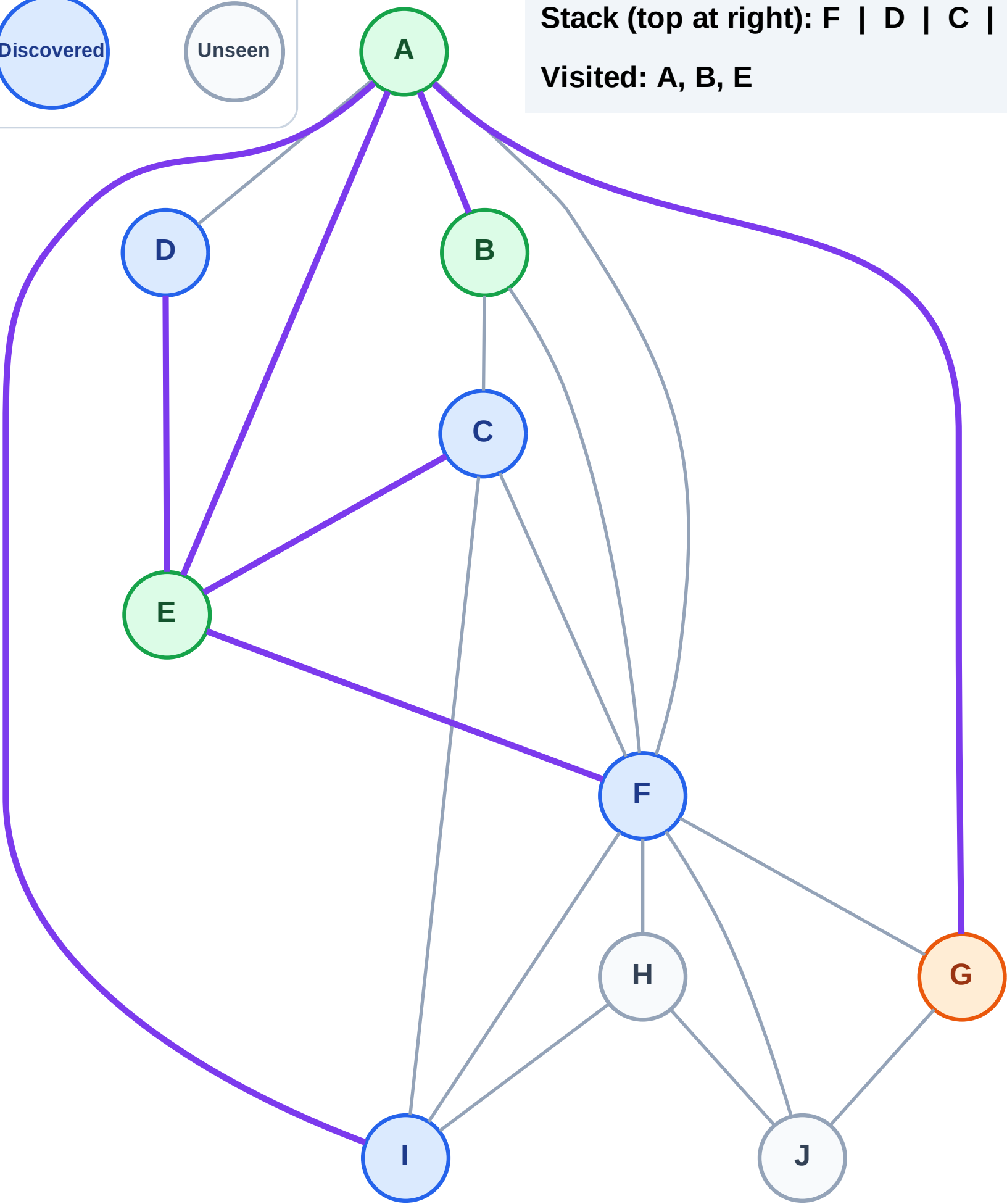
Discovered



Unseen

Stack (top at right): F | D | C | I

Visited: A, B, E



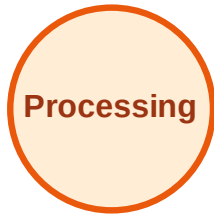
DFS Traversal

Step 9: Discover J from G

Legend



Complete



Processing



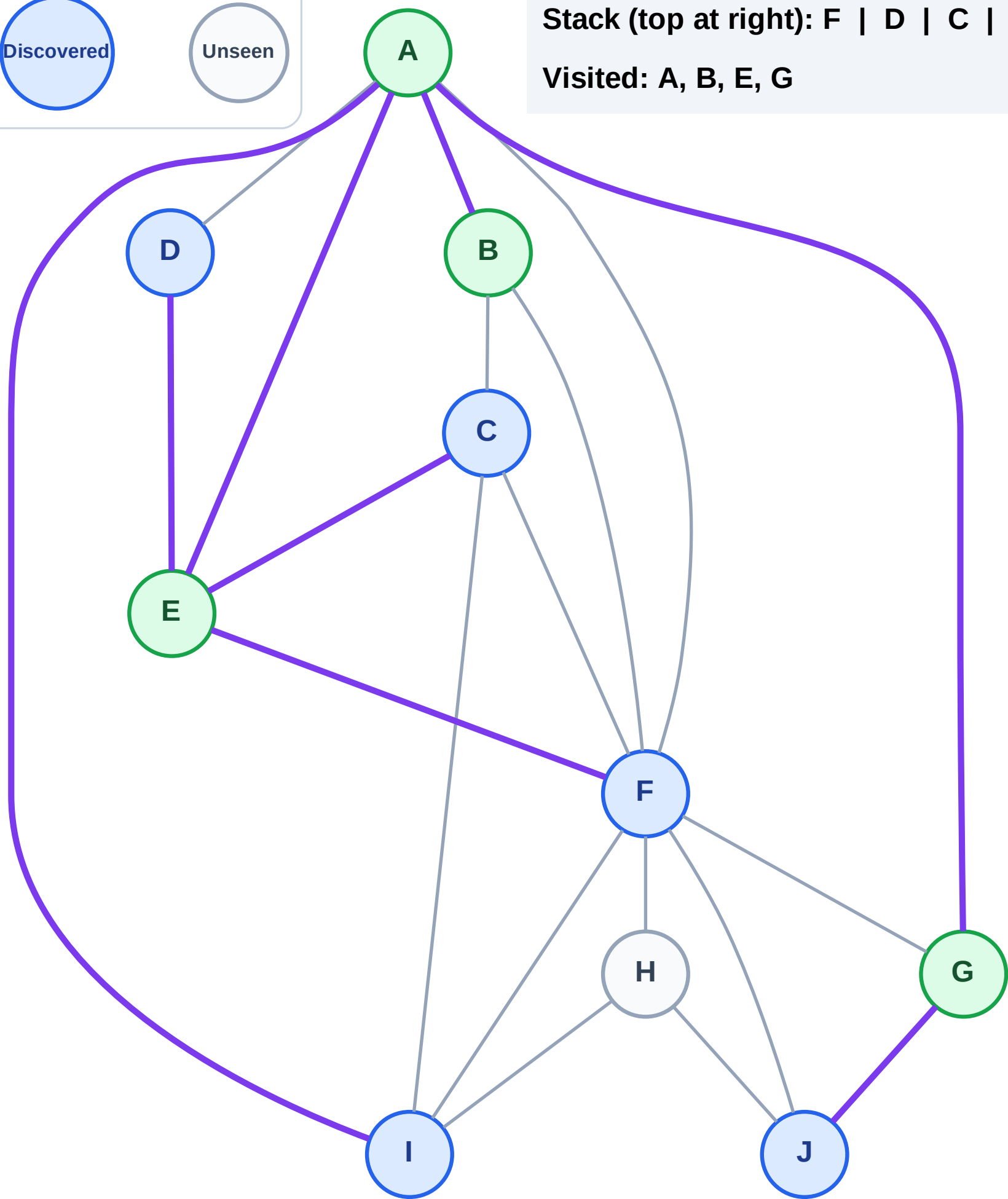
Discovered



Unseen

Stack (top at right): F | D | C | I | J

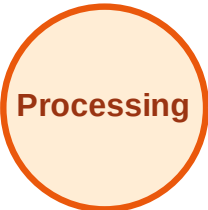
Visited: A, B, E, G



DFS Traversal

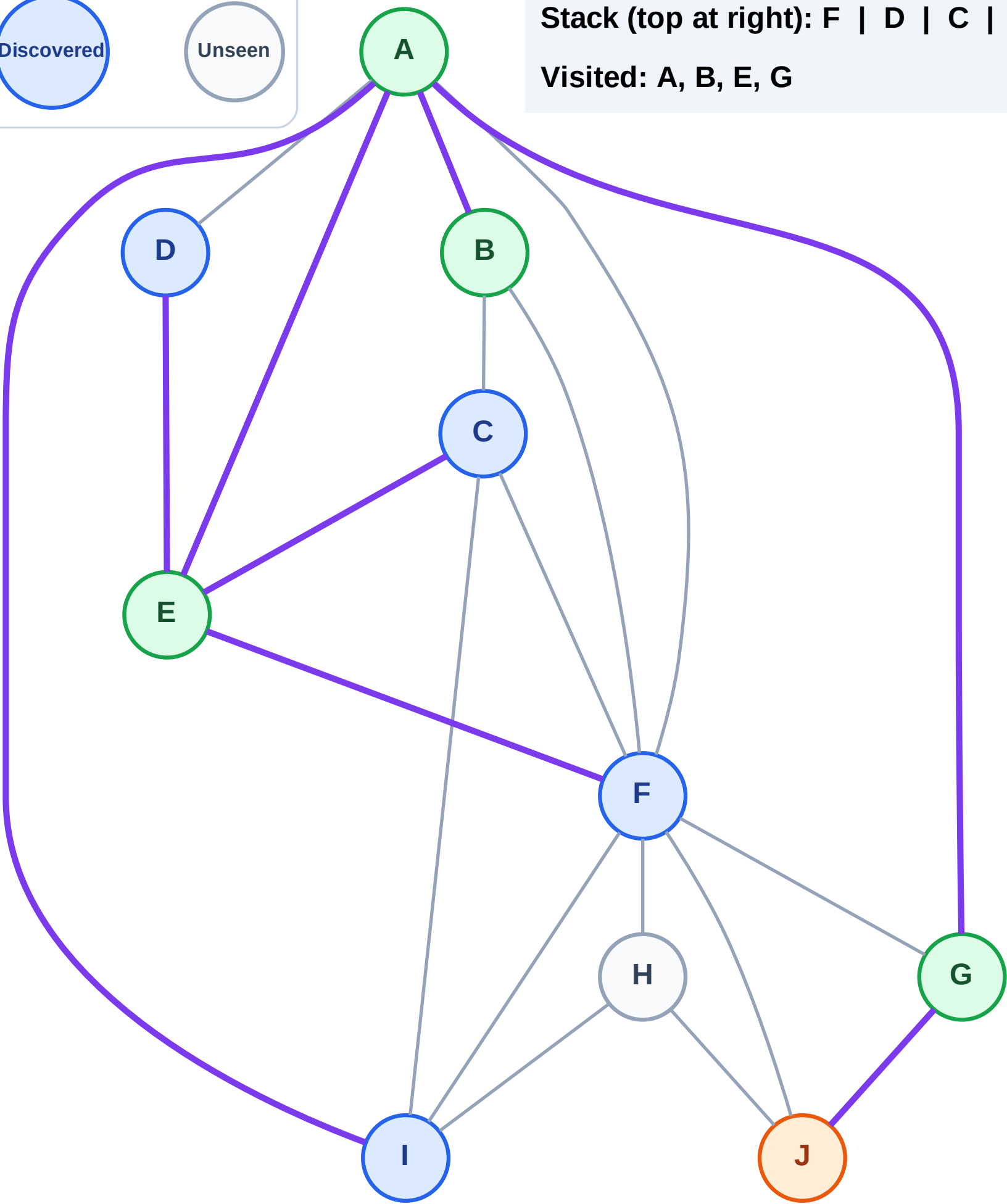
Step 10: Process node J

Legend



Stack (top at right): F | D | C | I

Visited: A, B, E, G



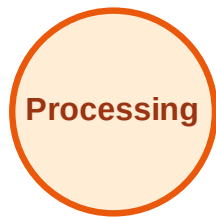
DFS Traversal

Step 11: Discover H from J

Legend



Complete



Processing



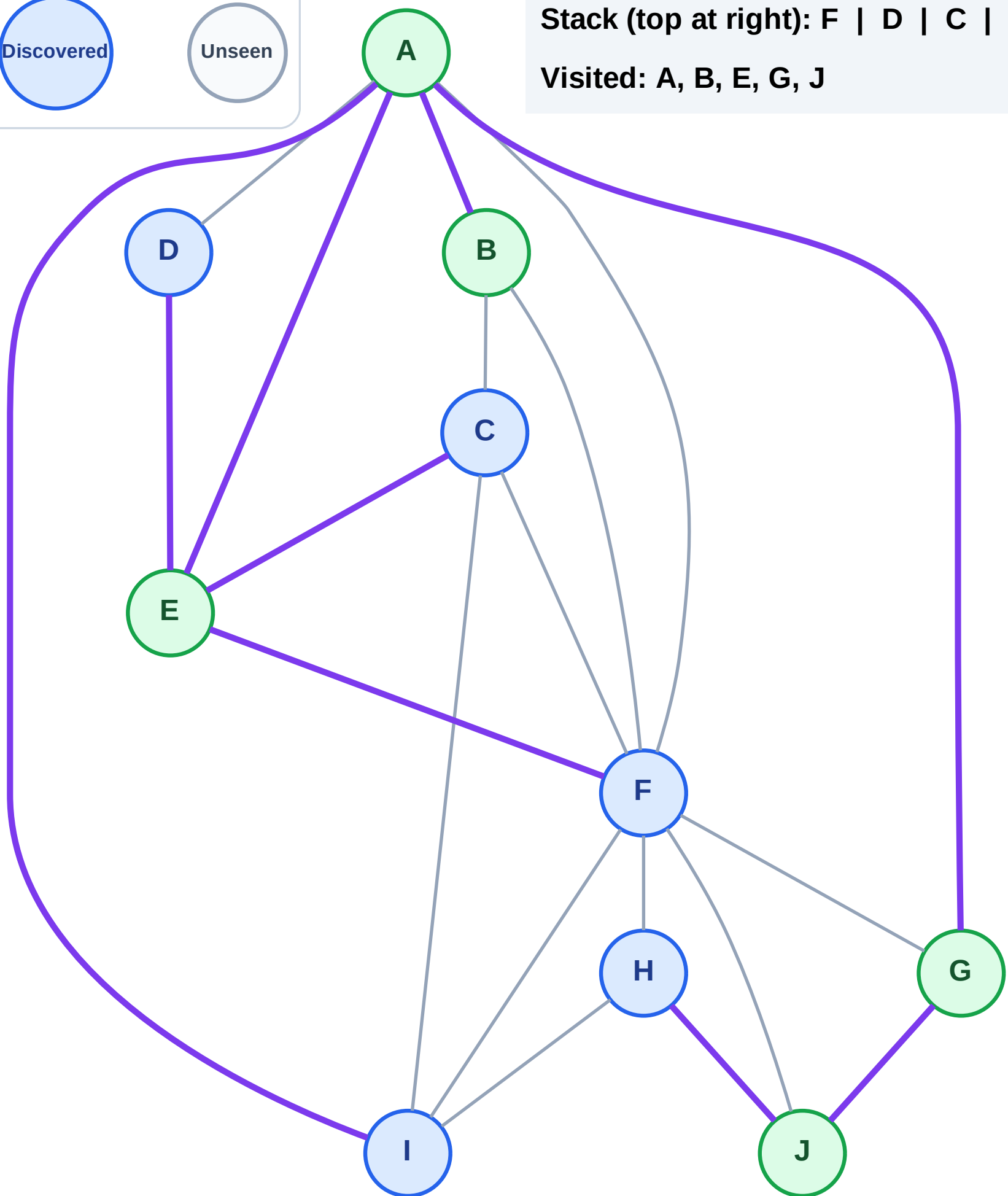
Discovered



Unseen

Stack (top at right): F | D | C | I | H

Visited: A, B, E, G, J



DFS Traversal

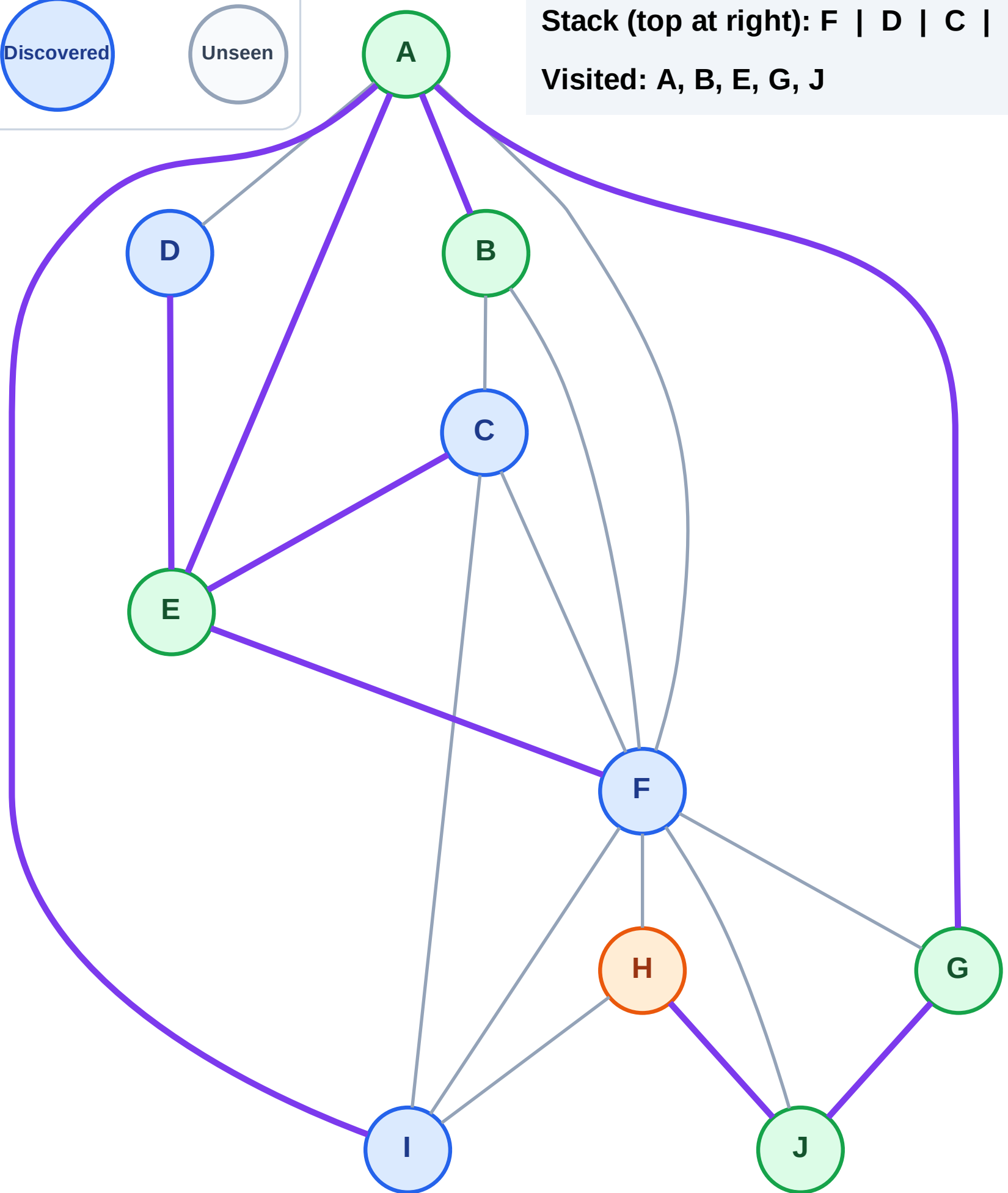
Step 12: Process node H

Legend



Stack (top at right): F | D | C | I

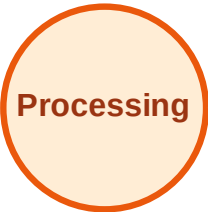
Visited: A, B, E, G, J



DFS Traversal

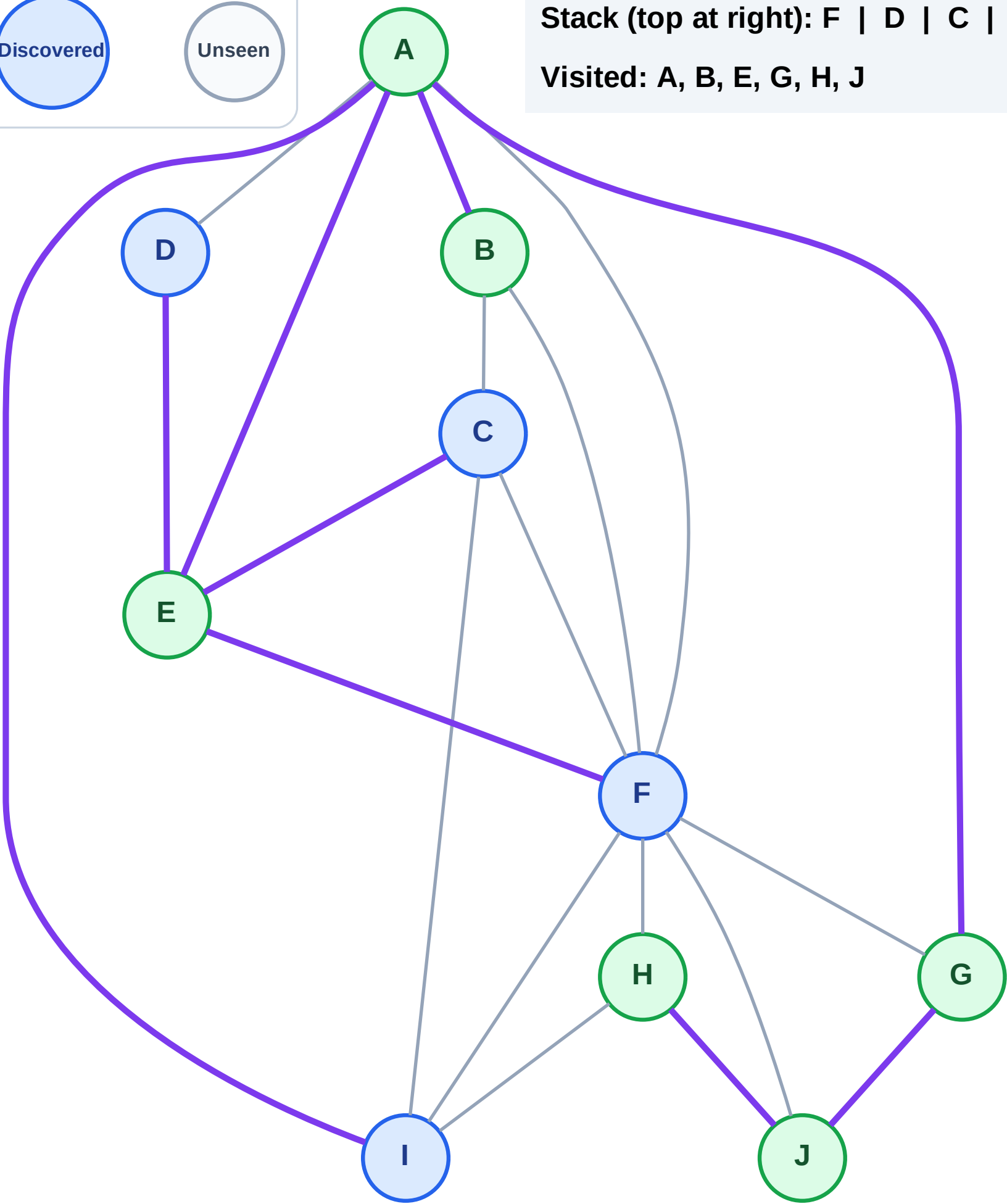
Step 13: No new neighbors from H

Legend



Stack (top at right): F | D | C | I

Visited: A, B, E, G, H, J



DFS Traversal

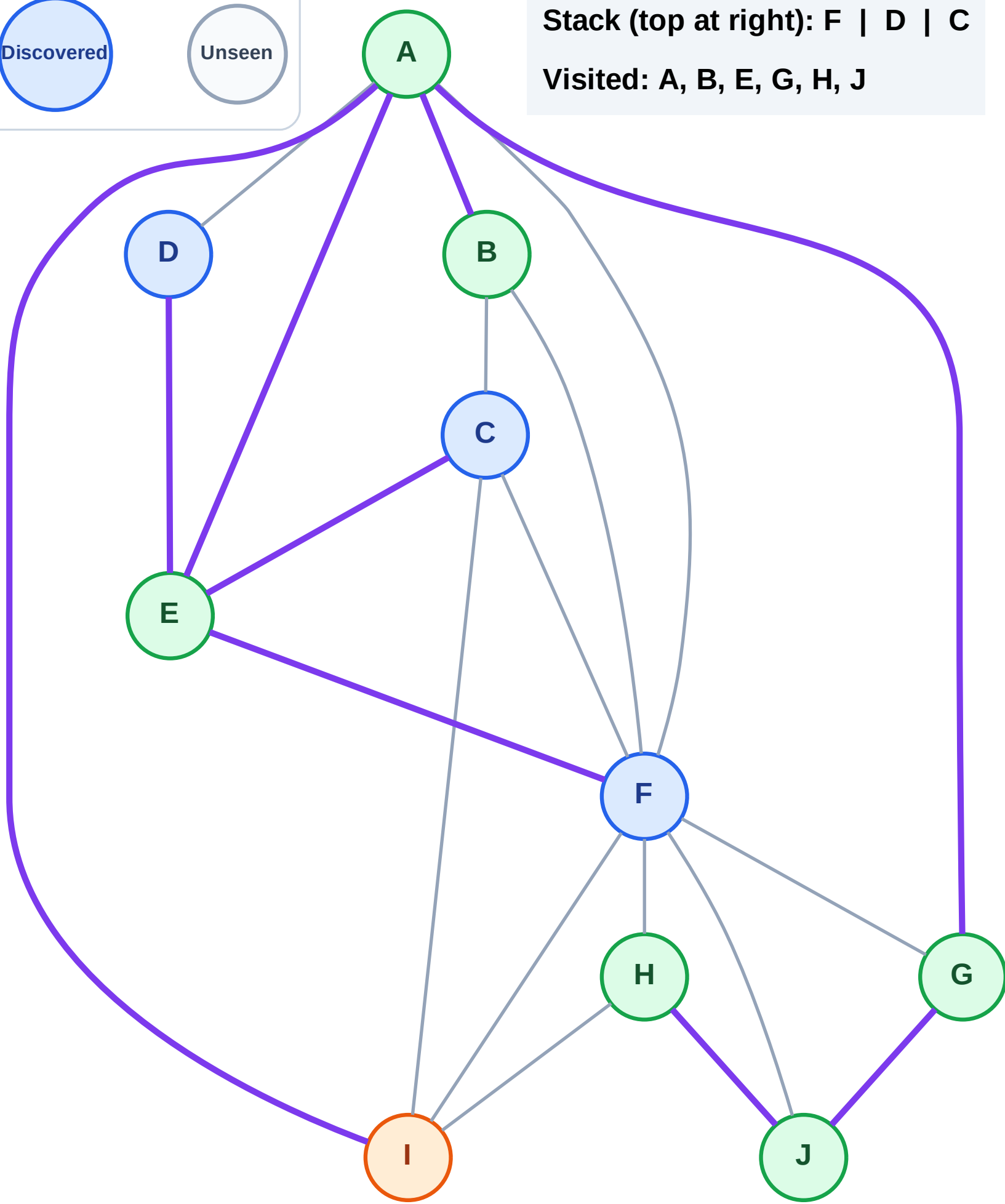
Step 14: Process node I

Legend



Stack (top at right): F | D | C

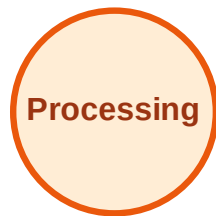
Visited: A, B, E, G, H, J



DFS Traversal

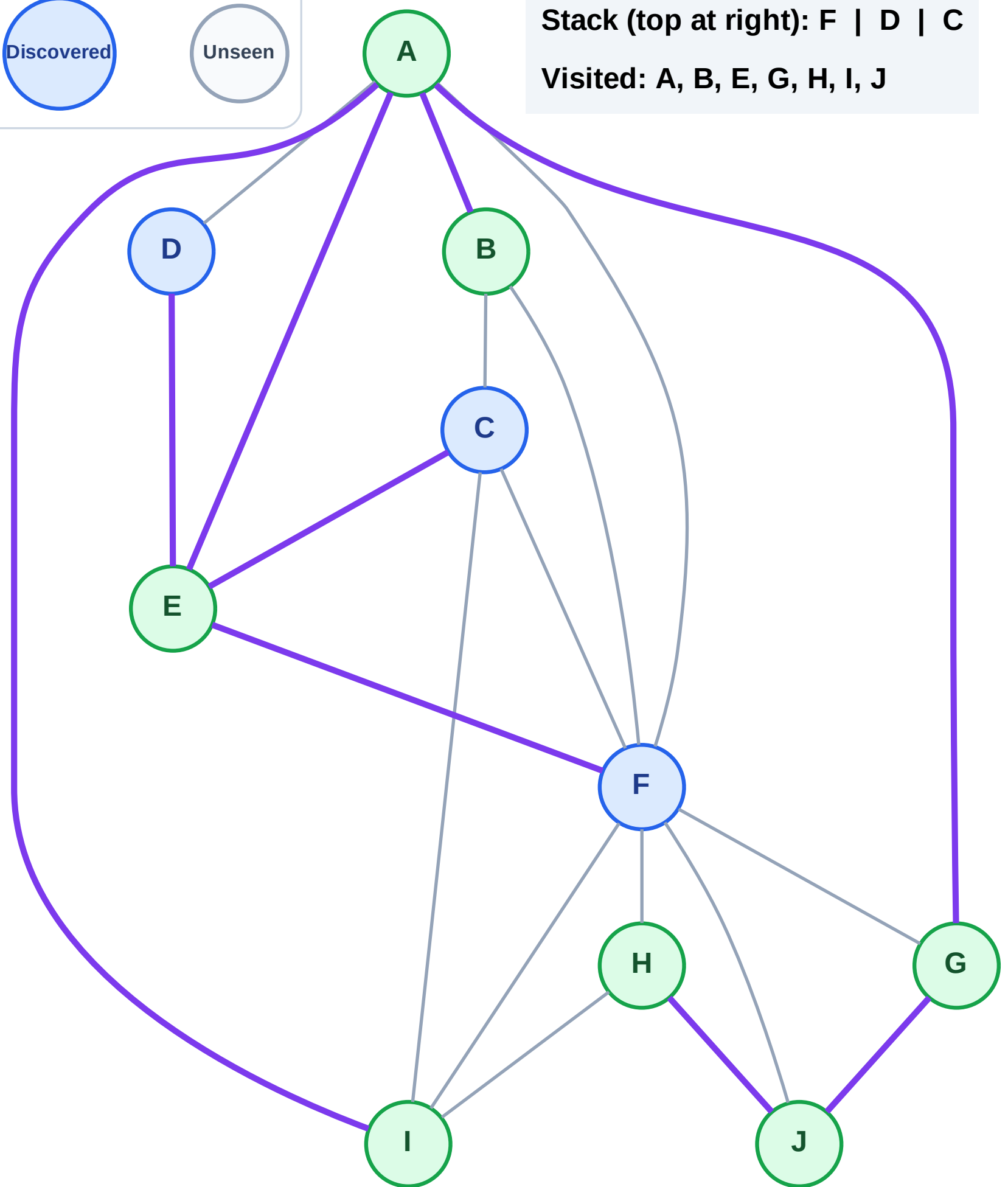
Step 15: No new neighbors from I

Legend



Stack (top at right): F | D | C

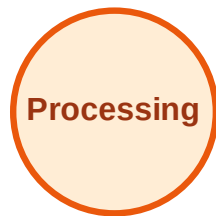
Visited: A, B, E, G, H, I, J



DFS Traversal

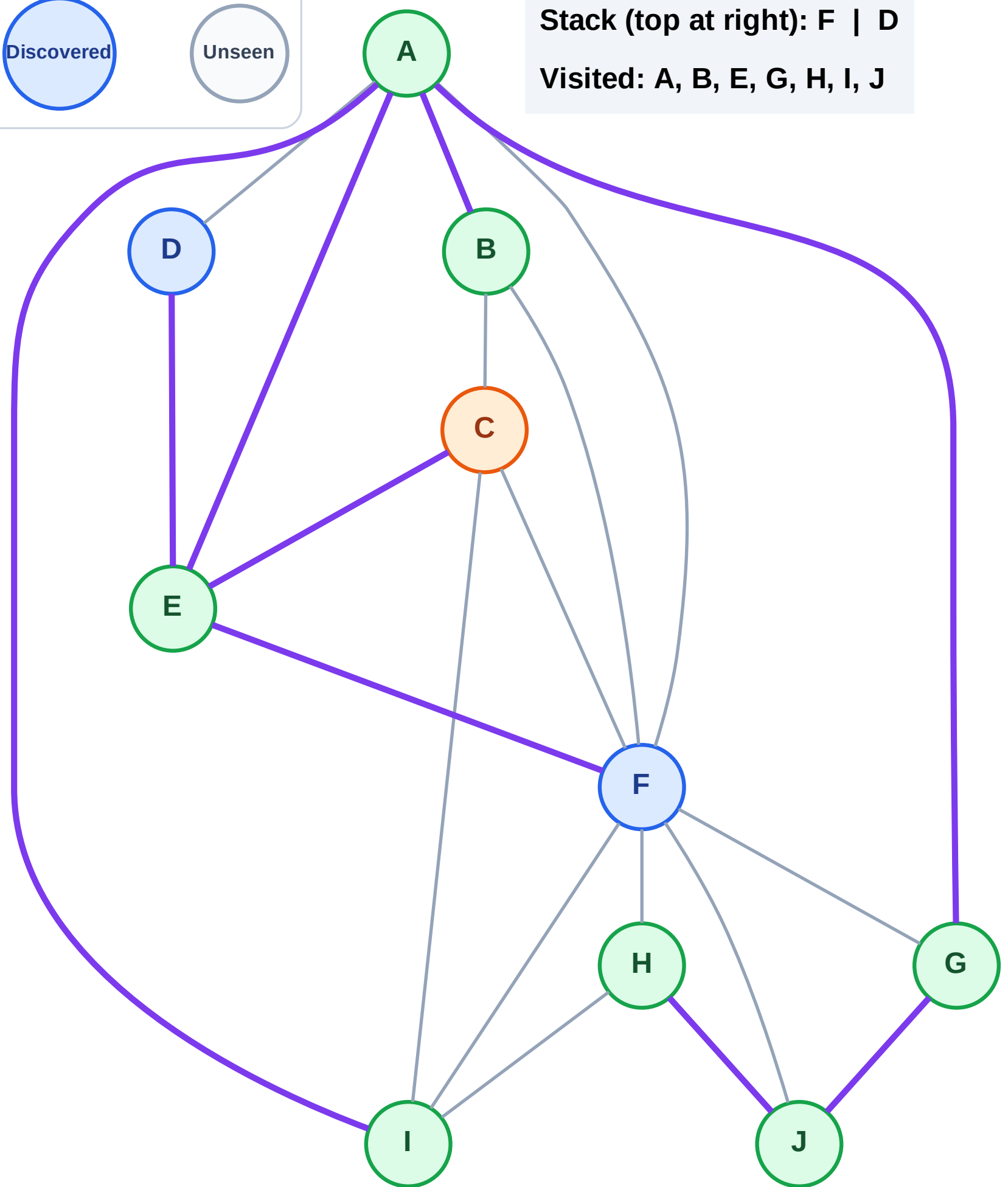
Step 16: Process node C

Legend



Stack (top at right): F | D

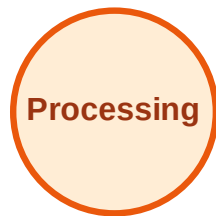
Visited: A, B, E, G, H, I, J



DFS Traversal

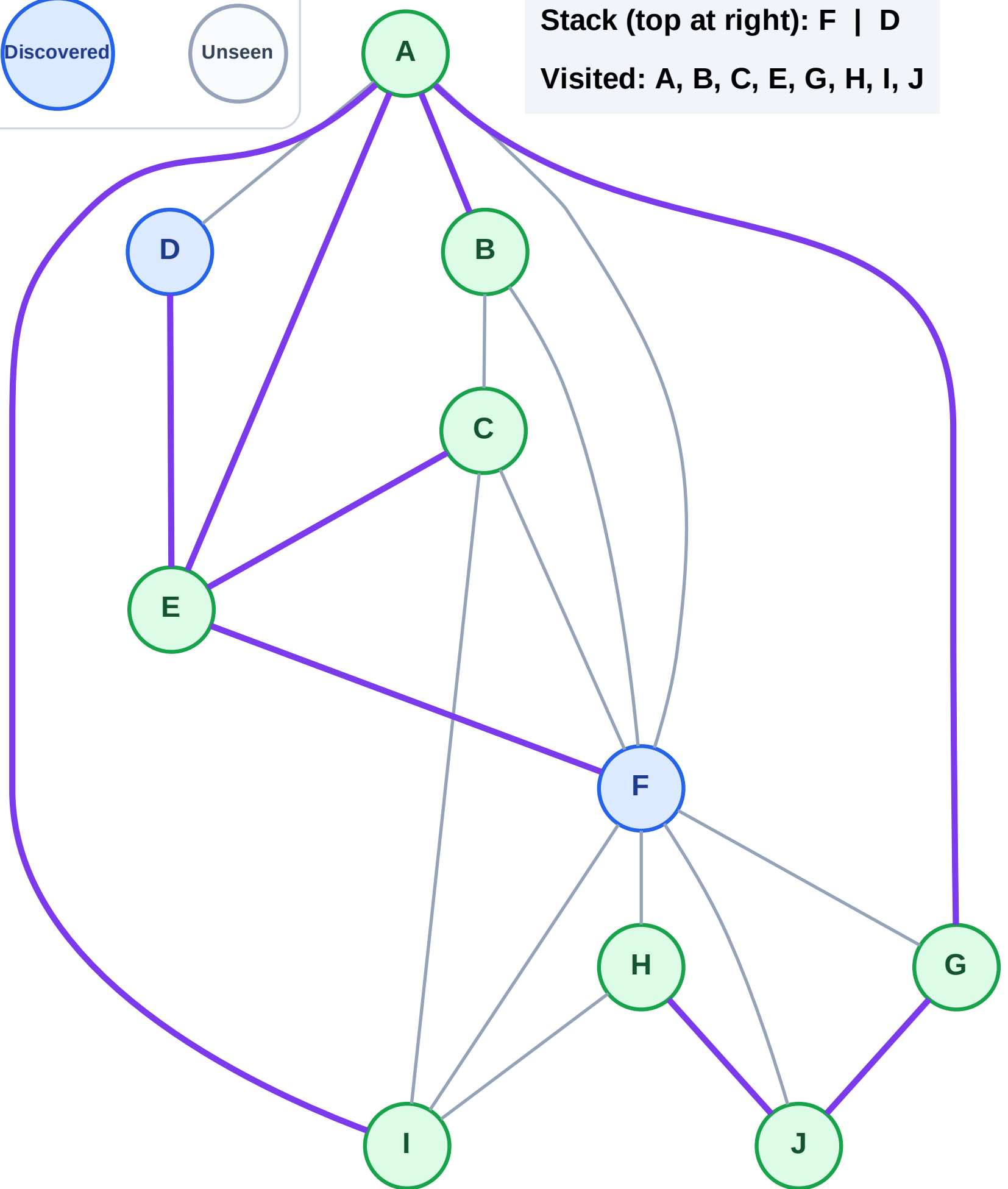
Step 17: No new neighbors from C

Legend



Stack (top at right): F | D

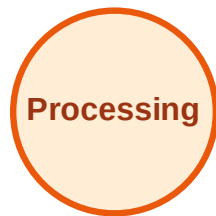
Visited: A, B, C, E, G, H, I, J



DFS Traversal

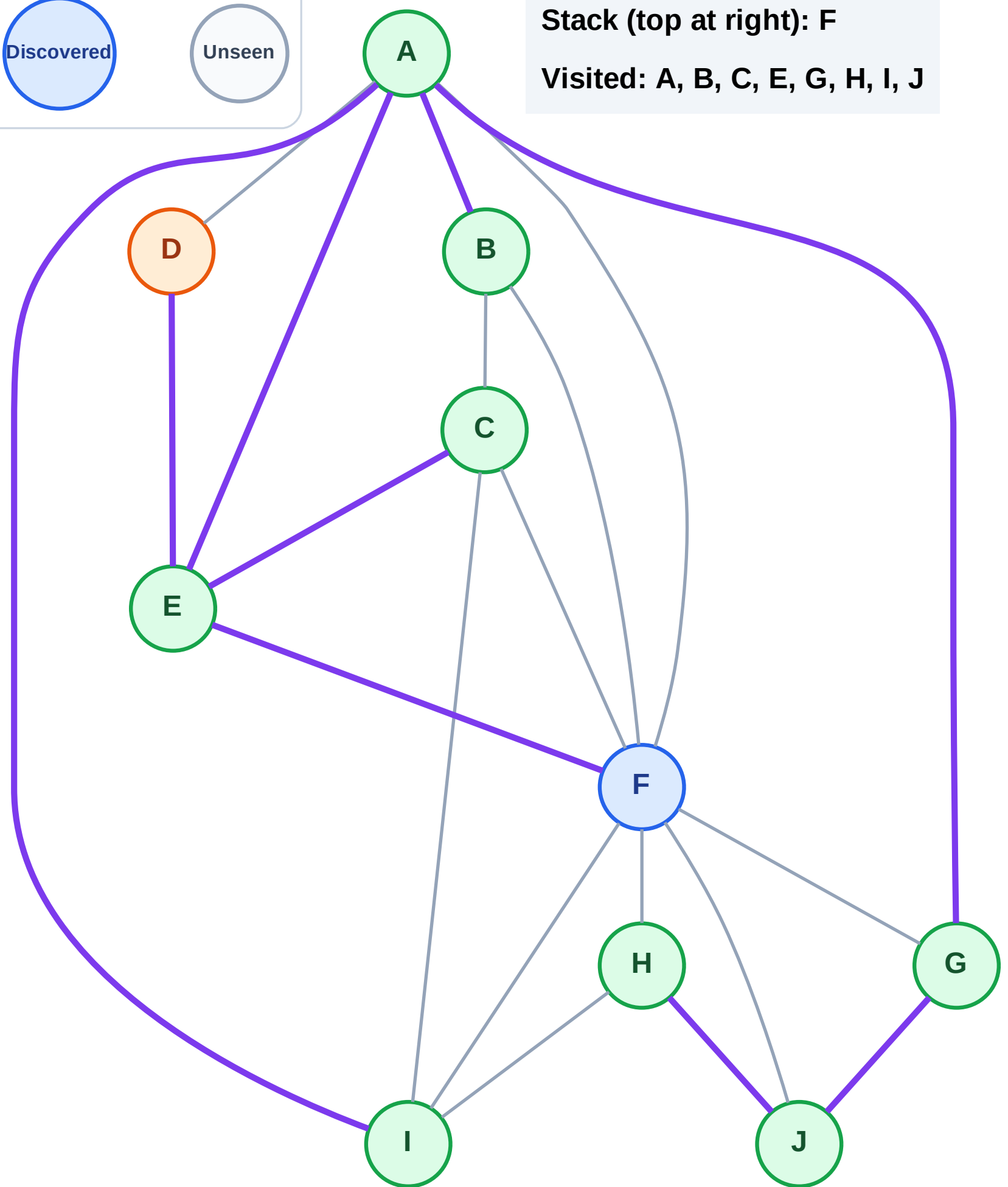
Step 18: Process node D

Legend



Stack (top at right): F

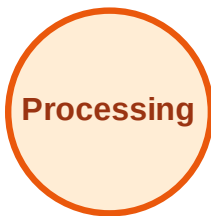
Visited: A, B, C, E, G, H, I, J



DFS Traversal

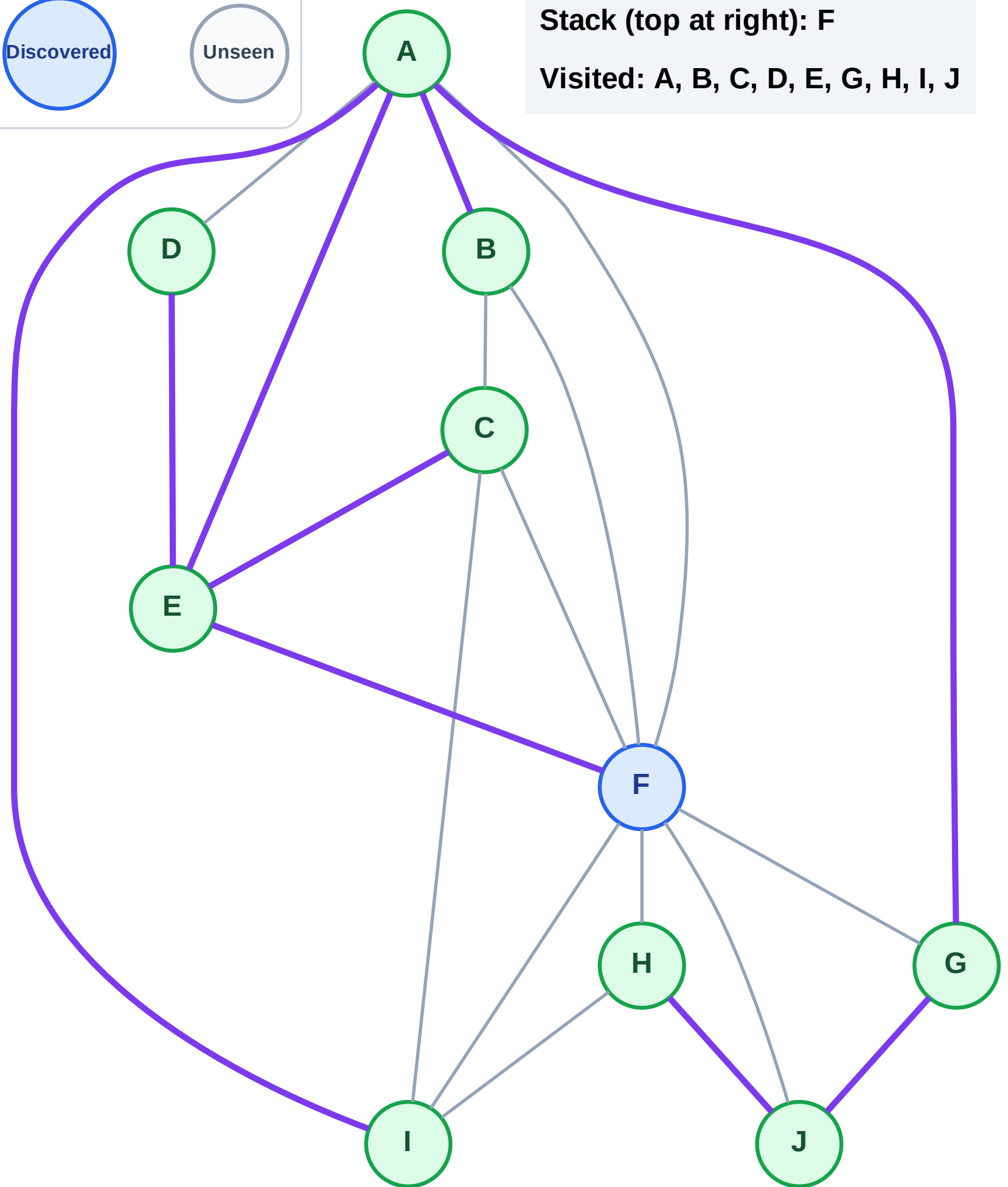
Step 19: No new neighbors from D

Legend



Stack (top at right): F

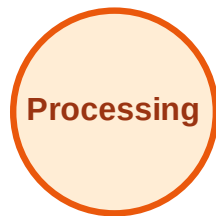
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

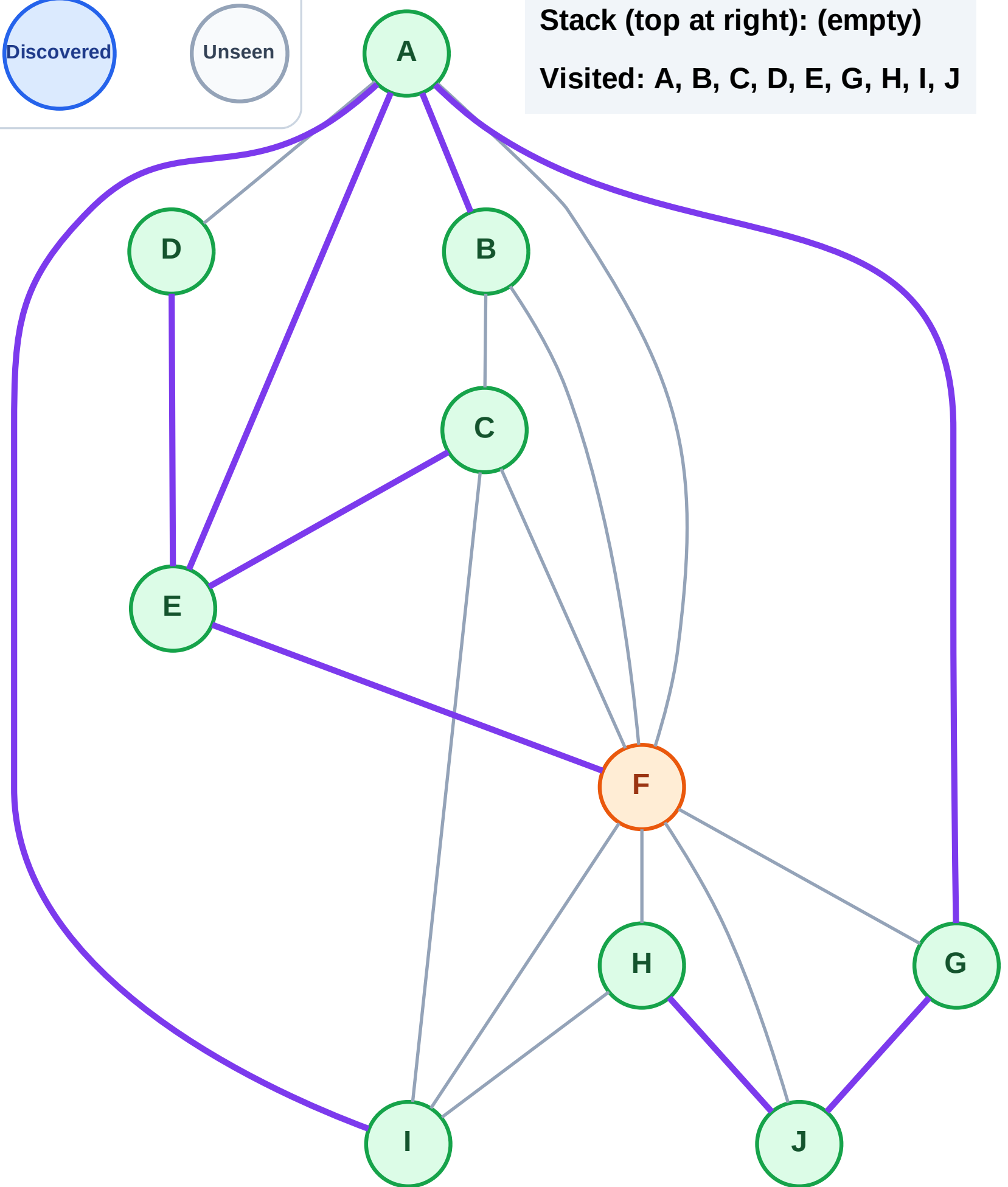
Step 20: Process node F

Legend



Stack (top at right): (empty)

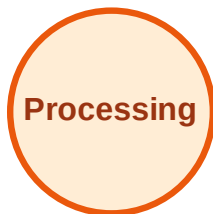
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

Step 21: No new neighbors from F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

